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MadStree: automated evaluation of tree-level de Sitter correlators

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Abstract: `MadStree` is a Mathematica package for the automated numerical evaluation of scalar cosmological correlators of arbitrary tree topology in de Sitter space. Starting from a diagram, it constructs the closed reduction system of the resulting family — integration-by-parts relations that reduce any integral of the family to master integrals — together with a d log-form differential equation and boundary data, then evaluates the master integrals along a user-specified kinematic path. Building on the general IBP and differential-equation formulas, we use additional relations of massless internal propagators to eliminate redundant master integrals while retaining closed reduction formulas and a d log-form system. Ordered large-energy limits provide boundary data for the resulting system, and the bundled `FlintNDE` backend shares local series between points to support efficient kinematic scans and analytic continuation. A common analytic regulator allows Laurent-series reconstruction when the time integrals diverge. We present massive and massless tree examples, comparisons with independent calculations, and an extension to the time integrals of loop graphs at fixed internal momenta.

Contents

1 Introduction **1**

2 IBP reduction and differential equations for tree-level dS integrals **3**
- 2.1 Notation: propagators, the integral family and index conventions 3
- 2.2 IBP relations from the three building blocks 8
- 2.3 d log-form differential equations 13
- 2.4 The remaining term and the contact blocks of the DE 15
- 2.5 Boundary conditions by nested blow-up 19
- 2.6 Several propagators sharing one theta 21

3 Using the package **23**
- 3.1 Installation and loading 23
- 3.2 Input formats and core functions 24
- 3.3 Numerical evaluation 32
- 3.4 Laurent-series reconstruction 35
- 3.5 Options, validation and common pitfalls 36

4 Examples **39**
- 4.1 Package examples 39
- 4.2 Application examples 40

5 Verification against independent references **55**

6 Summary and outlook **56**

A Common-theta bundles and the odd-subset expansion **57**

1 Introduction

Late-time correlation functions in inflation connect quantum field dynamics to the observed correlations of primordial fluctuations [1]. Their non-Gaussian structure can carry information about particles exchanged during inflation, including their masses and spins [2]. Trees with several massive scalar exchanges are relevant in quasi-single-field models, where they can dominate the three-point correlator [3]. Evaluating such diagrams over kinematic ranges is therefore useful for studying the momentum dependence of inflationary signals.

In de Sitter (dS) space, the in-in Feynman rules express scalar tree correlators as sums of time integrals containing Hankel functions, external-energy exponentials and time-ordering theta functions [1, 4]. Many analytic methods have been developed [2, 3, 5–37],

including the cosmological bootstrap, Mellin techniques and spectral decompositions. See also [38–43] for some recent progress in numerical methods.

We use a different approach, which is widely used in precision calculations in flat-spacetime QFT: Integration-By-Parts (IBP) relations [44] and the differential equations (DE) that they induce [45–48]. An IBP relation is a linear relation within an integral family. A sufficient set of these relations reduces the family to master integrals; differentiation and reduction then give a closed system of first-order DE. A d log-form connection [49] exposes its singularities and supports numerical transport from boundary data. In flat space, FIRE [50, 51] and Kira [52–54] automate IBP reduction, while generalized power series expansions [55], DiffExp [56] and the auxiliary mass flow method implemented in AMFlow [57–60] provide numerical evaluation. The determination of integrals from linear relations has also been developed in [61]. These methods motivate a workflow in which numerical values can be obtained without first finding a closed analytic expression.

The dS IBP/DE framework was developed in [42, 43, 62]. Reference [42] gives general iterative reduction formulas and d log-form DE for Hankel vertex integral families, including the contact terms needed for tree correlators. Reference [43] constructs power-series solutions, determines boundary data for two-vertex correlators through a blow-up of the multi-energy limit, and describes how the boundary factorization extends to general trees. These results provide the analytic basis for a systematic numerical implementation.

We develop this framework in two directions. First, we incorporate the extra algebraic relations of massless internal propagators into the general reduction and DE formulas. The generic Hankel formulas already apply to these integrals, but retain redundant mode-function states. For a time-ordered massless full propagator, the extra relations reduce four states to one shared two-state factor; a theta-free massless propagator contributes no state slot. We derive closed recurrences and a d log-form connection directly on this reduced space (Secs. 2.2 and 2.3). This reduces the size of the system to be transported without requiring a separate IBP solve for each diagram.

Second, we automate the reduction and the numerical evaluation for arbitrary tree families. From the tree specification, the package derives the closed reduction system of the family: every convergent integral — any contact sector and any integer shifts of the time powers — reduces to the master basis through closed recurrence formulas; the divergent configurations are handled by the analytic regulator described below. The time integrals of loop graphs at fixed internal momenta form families of the same kind and reduce the same way, extended by the common-theta contacts that cycles require (Sec. 2.6). Following the hierarchical energy limit of [43], we construct the boundary data for general tree topology: at leading order, the energy hierarchy fixes the theta functions and the resulting factorized integrals supply Gamma-function coefficients. Auxiliary energies include vertices without external legs in the same construction (Sec. 2.5). Boundary data and local solution series can then be shared among kinematic points. The benchmarks in Sec. 4.2 demonstrate the resulting multipoint efficiency and continuation beyond a direct series' convergence domain.

The package name `MadStree` stands for *Mad dS tree*. It implements both the reduced formulas and the numerical workflow in Mathematica, with `FlintNDE` as its bundled numer-

ical backend. For each specified Schwinger–Keldysh contour assignment, the input gives the tree topology, propagators, vertex energies and time powers. The package constructs the master system, the differential equation and boundary data, and returns values along the requested path; forming a physical correlator requires the contour sum and normalization factors appropriate to the model. The construction applies to the scalar Hankel and exponential integral families defined in Sec. 2.1.

Numerically, the same workflow covers these loop time integrals without performing the spatial loop-momentum integrations; the bubble example in Sec. 4.2 illustrates the extension. For time integrals that require regularization, the package can introduce a common analytic regulator on the vertex powers and determine the lowest Laurent power from the boundary and DE. It then reconstructs the coefficients numerically following the procedure used by AMFlow [59, 60]. The massless-chain example demonstrates the extraction of a pole coefficient and a finite part.

Sec. 2 presents the method, including the boundary construction, with the common-theta derivation in App. A. Sec. 3 describes package usage and Laurent reconstruction. Sec. 4 presents examples, and Sec. 5 compares the results with independent references. We conclude in Sec. 6.

2 IBP reduction and differential equations for tree-level dS integrals

This section constructs the reduced master space, its IBP recurrences and d log-form DE, and the boundary data needed for numerical evaluation. We first define the integral family and conventions, then derive the local blocks and assemble them over the graph. The massless relations are imposed before assembly, so the formulas act directly on the reduced space. Quantities inherited from the generic mode-function basis of [42] carry the superscript unreduced of (2.20). The boundary construction follows in Sec. 2.5. Finally, Sec. 2.6 extends the contact rules to several propagators sharing one theta function, as needed for time graphs with cycles.

2.1 Notation: propagators, the integral family and index conventions

In-in Feynman rules and asymptotic behavior. Cosmological correlators are evaluated in the Schwinger–Keldysh (SK) formalism: the operator product is ordered along a closed time path, and each vertex is labeled by a contour branch, written `"+"` or `"-"`, with one time integration $\int_{-\infty}^{0} \mathrm{d}\tau_v$ per vertex. With the positive-frequency mode function

$$u(\tau;k) = -\mathrm{i}\sqrt{\frac{\pi}{2}}\, e^{\mathrm{i}\pi(\nu/2+1/4)} H_{\text{Hubble}}^{(d-1)/2}(-\tau)^{d/2} H_\nu^{(1)}(-k\tau), \tag{2.1}$$

where $k = |\boldsymbol{k}|$ and $\nu^2 = d^2/4 - m^2/H_{\text{Hubble}}^2$ for a minimally coupled scalar in d spatial dimensions (ν is real or imaginary for $d = 3$), the bulk-to-bulk propagators are

$$G_>(k;\tau_1,\tau_2) \equiv u(\tau_1,k)u^*(\tau_2,k), \qquad G_<(k;\tau_1,\tau_2) \equiv u^*(\tau_1,k)u(\tau_2,k), \tag{2.2}$$

and the time-ordered combinations used by the in-in rules are

$$\begin{aligned} G_{++} &= G_{>}\theta(\tau_1-\tau_2)+G_{<}\theta(\tau_2-\tau_1), & G_{--} &= G_{<}\theta(\tau_1-\tau_2)+G_{>}\theta(\tau_2-\tau_1),\\ G_{+-} &= G_{<}, & G_{-+} &= G_{>}. \end{aligned} \tag{2.3}$$

Throughout, $\theta_{ij} \equiv \theta(\tau_i - \tau_j)$ denotes the theta function of the time difference of vertices i and j. Bulk-to-boundary propagators of massless fields, $G_{\pm} \propto (1 \mp \mathrm{i}k\tau)e^{\pm \mathrm{i}k\tau}$, enter only as pure-exponential blocks. In $d = 3$, a minimally coupled massless field has $\nu = 3/2$ and a conformally coupled one has $\nu = 1/2$; the two form one function family under the reduction, because the mode functions of every half-integer order are finite sums of exponentials multiplied by powers of τ, and the recurrences of Sec. 2.3 absorb the polynomial factors into shifted vertex time powers. The master integrals therefore fix the massless blocks at $\nu = 1/2$: every massless building block below carries this order. The Hankel functions obey

$$\partial_\tau^2 H_\nu^{(1,2)}(-k\tau) + \frac{1}{\tau}\,\partial_\tau H_\nu^{(1,2)}(-k\tau) + \left(k^2 - \frac{\nu^2}{\tau^2}\right) H_\nu^{(1,2)}(-k\tau) = 0, \tag{2.4}$$

with the two asymptotic regimes

$$\begin{aligned} &H_\nu^{(1)}(-k\tau) = c_1(-k\tau)^\nu(1+O(\tau^2)) + c_2(-k\tau)^{-\nu}(1+O(\tau^2)), \quad \tau \to 0^-,\\ &c_1 = e^{-\mathrm{i}\pi\nu}c[\nu]\,, \quad c_2 = c[-\nu]\,, \quad c[\nu] \equiv \frac{2^{-\nu}\Gamma(-\nu)}{\mathrm{i}\pi}\,. \end{aligned} \tag{2.5}$$

and

$$\begin{aligned} H_\nu^{(1)}(-k\tau) &\sim \sqrt{\frac{2}{\pi}}(-k\tau)^{-1/2}e^{-\mathrm{i}k\tau-\mathrm{i}\pi(\nu/2+1/4)},\\ H_\nu^{(2)}(-k\tau) &\sim \sqrt{\frac{2}{\pi}}(-k\tau)^{-1/2}e^{\mathrm{i}k\tau+\mathrm{i}\pi(\nu/2+1/4)}, \quad -k\tau \to \infty. \end{aligned} \tag{2.6}$$

The first regime fixes the boundary conditions of the time integrals; the second justifies the Wick rotation used in the numerical evaluation of Sec. 3.3.

The mode-function two-state h block. A tree-level correlator with c_V vertices contains one time integration per vertex, and each integrand is a product of the elementary factors

$$\left\{e^{\mathrm{i}k\tau},\ H_\nu^{(1,2)}(-k\tau),\ \partial_\tau H_\nu^{(1,2)}(-k\tau),\ \theta(\tau_j-\tau_k)\right\}. \tag{2.7}$$

As in [43], the Hankel prefactor and the time derivative are absorbed into two mode-function states of one function system. We use the convention of the package, in which the prefactor is the *positive* power of z,

$$\begin{aligned} &h^{(s)}(\nu,0;z) \equiv z^{+\nu}H_\nu^{(s)}(z),\\ &h^{(s)}(\nu,1;z) \equiv -\frac{1}{k}\,\partial_\tau h^{(s)}(\nu,0;z) = \partial_z h^{(s)}(\nu,0;z), \qquad z = -k\tau, \end{aligned} \tag{2.8}$$

where $s \in \{1,2\}$ selects the Hankel branch and $n \in \{0,1\}$ selects the undifferentiated function or its first derivative. A superscript on h always denotes the branch; when it is omitted, the statement applies to each branch separately. A massive propagator carries two independent mode-function state bits, one at each vertex. The relations of Sec. 2.2 reduce a massless full propagator (G_{++} or G_{--}) to one shared two-state slot. A theta-free massless propagator (G_{+-} or G_{-+}) contributes no state slot. The package works directly in these reduced spaces. The alternative normalization $h(\nu,0;z) = z^{-\nu}H_\nu(z)$ used in [42, 43] is related to (2.8) by $\nu \to -\nu$, because the reflection identities $H^{(1)}_{-\nu} = e^{\mathrm{i}\pi\nu}H^{(1)}_\nu$ and $H^{(2)}_{-\nu} = e^{-\mathrm{i}\pi\nu}H^{(2)}_\nu$ carry each branch into itself, multiplied only by a z-independent phase; the two phases are inverse to one another, so a normalization built from the pair of branches, such as eq. (2.48) below, is unaffected by the trade. Writing ν_{ref} for the order in those two papers, we set $\nu_{\mathrm{ref}} = -\nu$ throughout: in the equation of motion below, the Wronskian, M_1, the contact powers, the recurrences and the d log potentials. For a heavy field $\nu = \mathrm{i}\mu$ with real μ this reduces to $e^{-\pi\mu}H^{(2)}_{-\mathrm{i}\mu}(z) = H^{(2)}_{\mathrm{i}\mu}(z)$, and we shall use exactly this identity in Sec. 2.4 to trade the conjugate-order basis at the two vertices of a contracted propagator in [43] for the common-order basis of the package.

Since the Hankel equation involves ν^2 only, the two normalizations satisfy different first-order systems but the same second-order one up to $\nu \to -\nu$. In the convention (2.8), with $\boldsymbol{h} = (h(\nu,0;z), h(\nu,1;z))^T$ and $P_1 \equiv \frac{1}{2}(\mathbb{I} - \sigma_3) = \mathrm{diag}(0,1)$,

$$\partial_\tau \boldsymbol{h} = \left[-\mathrm{i}k\sigma_2 - \frac{1-2\nu}{\tau}\,P_1\right]\boldsymbol{h}, \tag{2.9}$$

where $\sigma_{1,2,3}$ are the Pauli matrices and $\mathbb{I}$ the 2×2 identity; the equivalent dimensionless equation of motion is

$$\partial_z^2 h(\nu,0;z) + \frac{1-2\nu}{z}\,\partial_z h(\nu,0;z) + h(\nu,0;z) = 0, \tag{2.10}$$

whose constant term is 1; the physical k^2 enters only through $\partial_\tau = -k\partial_z$.

The vertex integral family and its index conventions. Every integral in this paper is written with one symbol, I, whose arguments are the sector, the integer time-power shifts and the mode-function state bits. For a vertex carrying n massive legs the one-vertex family (an n-fold Hankel vertex integral family) reads

$$\mathrm{I}^{(a_0)}_{\tilde{\boldsymbol{a}};s_1\cdots s_n} = \int_{-\infty}^{0} \mathrm{d}\tau\,(-\tau)^{\nu_0+a_0} e^{\mathrm{i}p_0\tau} \prod_{i=1}^{n} h^{(s_i)}(\nu_i, a_i; -k_i\tau), \qquad a_0 \in \mathbb{Z}, \quad a_i \in \{0,1\}, \quad s_i \in \{1,2\}, \tag{2.11}$$

with the following index conventions, used throughout the graph:

- ν_0 is the power of the vertex time, $s_i \in \{1,2\}$ is the Hankel branch chosen at leg i by the SK type of the propagator that ends there, $a_i \in \{0,1\}$ is the *mode-function state bit* — the index of the two components of (2.8) — and a_0 is the integer *shift* of the vertex time power;

- $p_0 = \varsigma k_0$ is the signed energy of the vertex (Sec. 2.2), and $k_i = |\boldsymbol{k}_i|$ the momentum magnitude of leg i;
- the mode-function state bits are packed into one integer by the binary convention of [42],

$$\tilde{\boldsymbol{a}} = 1 + \sum_{i=1}^{n} a_i \, 2^{\,n-i}, \tag{2.12}$$

  so that the first bit is the most significant and the 2^n entries $\mathrm{I}_{\tilde{\boldsymbol{a}}} = \mathrm{I}_{\{\boldsymbol{a}\}}$ of the master vector are ordered exactly as the Kronecker products of the leg spaces; in the package this is `IntegerDigits[...,2,n]` and the same order is used for the whole graph.

For a graph the same three kinds of index are kept, only made global. Fix a sector s on a graph with c_V vertices; it has c_s time components, counted below, and n_s^{slot} mode-function two-state slots — one for each massive propagator at each of the two vertices it joins, and one shared by the two vertices of a massless full propagator — and write a master as

$$\mathrm{I}_{\tilde{\boldsymbol{r}}} \equiv \mathrm{I}_s(\boldsymbol{n}; \boldsymbol{a}), \qquad \boldsymbol{n} \in \mathbb{Z}^{c_s}, \quad \boldsymbol{a} \in \{0,1\}^{n_s^{\mathrm{slot}}}, \tag{2.13}$$

where $\boldsymbol{n}$ assigns one integer time-power shift to each component (a component is a maximal set of vertices merged by the contacts; a single vertex is one component), $\boldsymbol{a}$ is the ordered string of mode-function state bits of all slots, packed into $\tilde{\boldsymbol{r}}$ by (2.12), and the sector s itself is recorded by a *sector key*: with the root propagators listed in input order $E = (e_1, \ldots, e_{N_E})$ and L_s^{ctr} the set of contracted propagators,

$$\mathrm{key}(s) = \xi_1 \xi_2 \cdots \xi_{N_E}, \qquad \xi_j = \begin{cases} 0, & e_j \in L_s^{\mathrm{ctr}}, \\ 1, & e_j \notin L_s^{\mathrm{ctr}}, \end{cases} \tag{2.14}$$

a fixed-length bit *string*, not a binary integer, so leading zeros are kept and the top sector is the all-1 string; every other sector contains at least one contracted propagator and is called *pinched*. Each contracted propagator merges two components and removes one time integration, and the graph is a tree, so

$$c_s = c_V - \left| L_s^{\mathrm{ctr}} \right|, \tag{2.15}$$

and only the top sector has as many shifts as the graph has vertices. Sectors are partially ordered by $t \prec s$ when t is reachable from s by at least one contact, i.e. when $L_t^{\mathrm{ctr}} \supsetneq L_s^{\mathrm{ctr}}$. This ordering is the tree-level counterpart of the *subsector* relation of flat-space IBP, where S is a subsector of S' if every propagator power of S is less than or equal to the corresponding power of S' [53]: the integer propagator powers that order flat-space sectors are here replaced by the contact indicator bits ξ_j, since a propagator of a tree is either kept ($\xi_j = 1$) or contracted away ($\xi_j = 0$).

These three indices also label the package object `MSIntegral[sectorKey, aShifts, stateBits]`, which represents the normalized integral $J_s = \mathcal{N}_s \widehat{\mathrm{I}}_s$ defined below. Here `sectorKey` is the string (2.14), `aShifts` the vector $\boldsymbol{n}$ and `stateBits` the string $\boldsymbol{a}$; the

same slots are used by the returned records `"sectorKey", "stateBits", "integral", "normalization"`. The object maps slot-by-slot onto the corresponding `dSIBP` object `J[sectorKey, timeShifts, stateBits]` of the parent framework [63], with the same key alphabet, the same shift dimension c_s and the same bit order, so that a `MadStree` master can be read as a `dSIBP` J and vice versa without any re-indexing.

Suppressed data and normalization. The indices in (2.13) label the integral within a fixed context. Its full definition also includes the following data:

1. the Hankel prefactor $z^{\pm\nu}$ and the normalization of the mode function (2.1): the $(-\tau)^{d/2}$ and the $z^{\pm\nu}$ powers are split between the explicit vertex power ν_0 and the definition (2.8), so no propagator carries a visible prefactor;

2. the choice of the small-τ branch and of the vanishing powers at $\tau \to 0$: both are contained in the superscript $(1, 2)$ and the order ν of $h^{(s)}$, which select between the two terms of (2.5);

3. the sector normalization $\mathcal{N}_s$: the object the package propagates is $J_s(\boldsymbol{n}; \boldsymbol{a}) = \mathcal{N}_s \widehat{\mathrm{I}}_s(\boldsymbol{n}; \boldsymbol{a})$, where $\widehat{\mathrm{I}}_s = Q_s \mathrm{I}_s$ includes the massless basis change defined in (2.28), with
$$\mathcal{N}_s = \mathcal{N}_s^{(0)} \prod_{\substack{e \in L_s^{\mathrm{ctr}} \\ \nu_e \neq 1/2}} \left(-\sigma_e \, \frac{4\mathrm{i}}{\pi} \, q_e^{2\nu_e - 1} \right), \tag{2.16}$$
one factor per contracted massive full propagator, the constant that Sec. 2.4 attaches to the contact term (2.49); ν_e is the order and q_e the momentum magnitude of that propagator, and σ_e its contour sign, $+1$ for G_{++} and -1 for G_{--}; a contracted massless full propagator contributes the factor 1, and $\mathcal{N}_s^{(0)}$ is the normalization of the sector seed, equal to 1 for every example of Sec. 4;

4. the SK contour signs: the vertex phase sign ς_v of (2.17), the incidence ϵ_{ve} of (2.29) and the propagator sign σ_e of (2.27) are fixed by the vertex types and by the vertex order of each propagator, and enter only through the signed energies $p_{0,v}$ and the letters κ;

5. the theta chamber: at a boundary chart each theta is frozen to 0 or 1 by the strict energy order (Sec. 2.5), and inside the bulk the ordering of a bundle is carried by the single δ of Sec. 2.6;

6. the regulator: when a time power sits on a pole, a common analytic shift $\nu_0 \to \nu_0 + \varepsilon$ is understood (Sec. 3.4).

These data determine the coefficients and boundary conditions even though they are suppressed in the integral label. In particular, the momentum dependence of $\mathcal{N}_s$ contributes to the normalized connection, as described in Sec. 2.4.

2.2 IBP relations from the three building blocks

The building blocks assembled in this subsection drive the reduction: from them, the one-step relations of Sec. 2.3 express an integral whose time power at one vertex is shifted by one unit in terms of integrals one step closer to the master value, and their iteration reduces shifted integrals of the family to the zero-shift masters. Three building blocks contribute — the theta-free vertex exponential, the massive vertex h block and the massless shared block; their relative normalization is fixed first.

The relative normalization of the recurrences. The recurrences below use the $+\nu$ convention and the $(-\tau)^{\nu_0}$ time power of (2.11). With $(-\tau)^A$ instead of τ^{ν_0}, the derivative of the explicit power picks up a sign, $\partial_\tau(-\tau)^A = -A(-\tau)^{A-1}$, so the one-step relation carries $-\mathrm{M}_1$ where the reference carries $+\mathrm{M}_1$; the diagonal part of the DE is unaffected but the contact block of the DE changes sign, which is why our (2.59) differs from the printed [43] form by that overall sign.

Block (i): the theta-free exponential of a vertex. Each vertex v carries exactly *one* theta-free exponential factor,

$$E_v(\tau_v) = \exp(\mathrm{i}\varsigma_v k_{0,v}\tau_v)\,, \qquad \varsigma_v = \begin{cases} -1, & v \text{ is a } + \text{ vertex}, \\ +1, & v \text{ is a } - \text{ vertex}, \end{cases} \qquad p_{0,v} = \varsigma_v k_{0,v}, \tag{2.17}$$

so that a `"+"` vertex gives $e^{-\mathrm{i}k_{0,v}\tau_v}$ and a `"-"` vertex gives $e^{+\mathrm{i}k_{0,v}\tau_v}$. Here $k_{0,v}$ is the *single summed* external-energy parameter of the vertex, the sum of the magnitudes of the theta-free external legs attached to v; in the package it is the input field `externalLegEnergy`. Eq. (3.1) of [42] likewise writes one k_0 per vertex, with no separate exponential parameters for individual legs or blocks. A theta-free propagator branch is not folded into this symbol: it keeps its own magnitude, and that magnitude is added to the energy of whichever component holds the vertex it ends on. After a contact has merged a set C of vertices into one component, the same scalar becomes the component energy

$$p_{0,C} = \sum_{v\in C} \varsigma_v k_{0,v} + \sum_{\substack{\text{theta-free propagators } e \\ \text{with exactly one vertex in } C}} (-\varsigma_{v_e})\, q_e, \tag{2.18}$$

where $q_e = |\boldsymbol{q}_e|$ is the momentum magnitude of the propagator and v_e is the vertex of e that lies in C; a branch whose two vertices both lie in C contributes $+q_e - q_e = 0$ and is therefore already absent from this sum. The second sum collects the massless cross and external branches, which create no slot: they add only their magnitudes to the same scalar, each with the contour sign of the vertex where the branch ends. The first sum alone is $p_{0,v}$, so for a one-vertex component (2.18) reduces to $p_{0,v}$ exactly when that vertex carries no theta-free propagator branch.

Writing $\mathcal{E}_v \simeq \mathbb{C}$ for the one-dimensional space of this exponential, the block creates no mode-function state bit and no slot: it contributes nothing to M_1 and contributes the scalar $\mathrm{i}p_{0,v}\mathbb{I}$ to M_0.

Block (ii): the h system of a massive propagator. A massive propagator of order ν contributes, at each of the two vertices it joins, one mode-function two-state system (2.8), with the two-dimensional space $\mathcal{H}_i \simeq \mathbb{C}^2$ — one factor per propagator end, indexed by i in the graph below — and the generators (2.9). This is the block of [42] unchanged, with $1-2\nu$ replacing the printed $2\nu+1$. Only the branch $s \in \{1,2\}$, fixed by the SK type of the propagator, differs between the two vertices it joins, and it does not change the generator $-\mathrm{i}k\sigma_2$.

Block (iii): the massless full propagator. Before the extra massless relations are imposed, a massless full propagator $e = (u_e, v_e)$ is exactly like a massive one: the vertex u_e carries two mode-function states $a \in \{0,1\}$, the vertex v_e carries two mode-function states $b \in \{0,1\}$, and the local space of this propagator is $\mathcal{H}_{e,u} \otimes \mathcal{H}_{e,v} \simeq \mathbb{C}^4$ with, in the natural order,

$$\mathrm{I}_e^{(4)} = (\mathrm{I}_{00}, \mathrm{I}_{01}, \mathrm{I}_{10}, \mathrm{I}_{11})^T, \tag{2.19}$$

where the I_{ab} are the integrals defined by (2.22) below, in the convention (2.13). The graph-level matrices follow from the Leibniz rule. The integrand is a product of independent factors, one exponential per vertex and one first-order mode-function system per propagator end, so ∂_{τ_v} differentiates one factor at a time and every other factor is a spectator. Bookkeeping the states as a tensor product, with $\mathrm{Emb}_i(A) = \mathbb{I} \otimes \cdots \otimes A \otimes \cdots \otimes \mathbb{I}$ acting on the slot i alone, each vertex matrix is the sum of the embedded generators of the factors that depend on τ_v:

$$\begin{aligned}
\mathcal{V}_s^{\text{unreduced}} &= \bigotimes_{v \in V_s} \mathcal{E}_v \otimes \bigotimes_{\substack{\text{active two-state} \\ \text{factors } i}} \mathcal{H}_i, \qquad \dim \mathcal{V}_s^{\text{unreduced}} = 2^{N_s}, \\
\mathrm{M}_{1;v}^{\text{unreduced}} &= \nu_{0,v}\mathbb{I} - \sum_{i \in H_v^{\text{unreduced}}} (1-2\nu_i)\,\mathrm{Emb}_i(P_1), \\
\mathrm{M}_{0;v}^{\text{unreduced}} &= \mathrm{i}p_{0,v}\mathbb{I} - \mathrm{i} \sum_{i \in H_v^{\text{unreduced}}} k_i\,\mathrm{Emb}_i(\sigma_2),
\end{aligned} \tag{2.20}$$

Here V_s is the set of active vertices of s, $H_v^{\text{unreduced}}$ the mode-function two-state factors of the propagators ending on v — massive and massless alike, before the reduction below — and N_s their total number in s; the product carries one $\mathcal{E}_v$ per active vertex and one $\mathcal{H}_i$ per two-state factor. These are the matrices of [42] in our convention, with the $+\nu$ prefactor (2.8) and the $(-\tau)^{\nu_0}$ time power.

Imposing the extra massless relations reduces these four mode-function states at the two vertices to one shared mode-function two-state system. At the conformal value $\nu = 1/2$ the positive-prefactor block degenerates: the two branches are pure exponentials, $h^{(1)}(\frac{1}{2}, 0; z) = -\mathrm{i}\sqrt{2/\pi}\, e^{\mathrm{i}z}$ and $h^{(2)}(\frac{1}{2}, 0; z) = +\mathrm{i}\sqrt{2/\pi}\, e^{-\mathrm{i}z}$, (2.10) becomes $h'' + h = 0$, and the derivative mode-function state is a branch eigenstate of the zeroth-order mode-function

state. Carrying the derivative out explicitly,

$$\begin{aligned} h^{(1)}(\tfrac{1}{2},1;z) &= \partial_z\left(-\mathrm{i}\sqrt{\tfrac{2}{\pi}}\,e^{\mathrm{i}z}\right) = \sqrt{\tfrac{2}{\pi}}\,e^{\mathrm{i}z} = +\mathrm{i}\,h^{(1)}(\tfrac{1}{2},0;z), \\ h^{(2)}(\tfrac{1}{2},1;z) &= \partial_z\left(+\mathrm{i}\sqrt{\tfrac{2}{\pi}}\,e^{-\mathrm{i}z}\right) = \sqrt{\tfrac{2}{\pi}}\,e^{-\mathrm{i}z} = -\mathrm{i}\,h^{(2)}(\tfrac{1}{2},0;z). \end{aligned} \tag{2.21}$$

Consistently with the fixed massless order of Sec. 2.1, the quotient derived here acts on the pure exponential blocks: an input propagator of another half-integer order must first be expanded into $\nu = 1/2$ blocks with shifted vertex powers. Its defining property is that the two branches carry *opposite* signs in (2.21), while a single *full* propagator puts the two branches on its two vertices in both possible orders, one per theta term. Write $\sigma_e = +1$ for G_{++} and $\sigma_e = -1$ for G_{--}. For a massless full propagator $e = (u_e, v_e)$ of momentum magnitude q_e, with $\tau_u \equiv \tau_{u_e}$ and $\tau_v \equiv \tau_{v_e}$, the four entries of (2.19) are, for both G_{++} and G_{--},

$$\mathrm{I}_{00} = \mathrm{I}_{11} = A_e + B_e, \qquad \mathrm{I}_{10} = -\mathrm{I}_{01} = \mathrm{i}\sigma_e\,(A_e - B_e)\,, \tag{2.22}$$

in terms of the two theta chambers of the propagator, taken undifferentiated,

$$\begin{aligned} A_e &= \int_{-\infty}^{0}\mathrm{d}\tau_u \int_{-\infty}^{0}\mathrm{d}\tau_v\; \Phi_e\,\theta_{uv}\,h^{(1)}\big(\tfrac{1}{2},0;-q_e\tau_u\big)\,h^{(2)}\big(\tfrac{1}{2},0;-q_e\tau_v\big)\,, \\ B_e &= \int_{-\infty}^{0}\mathrm{d}\tau_u \int_{-\infty}^{0}\mathrm{d}\tau_v\; \Phi_e\,\theta_{vu}\,h^{(2)}\big(\tfrac{1}{2},0;-q_e\tau_u\big)\,h^{(1)}\big(\tfrac{1}{2},0;-q_e\tau_v\big)\,, \end{aligned} \tag{2.23}$$

which is (2.3) written out in the h mode-function state basis: the branch pair of each chamber is the one carried by G_{σ_e}, so (2.23) displays G_{++}, and G_{--} exchanges the two Hankel branches inside each term. Here Φ_e collects every other factor of the integrand: the vertex exponentials of u and v, their time powers, and the factors of all the remaining propagators. The relations (2.22) follow because raising the mode-function state at u_e from 0 to 1 multiplies A_e by $+\mathrm{i}\sigma_e$ and B_e by $-\mathrm{i}\sigma_e$, and raising it at v_e multiplies them by $-\mathrm{i}\sigma_e$ and $+\mathrm{i}\sigma_e$: the sign of the eigenvalue (2.21) is fixed by the branch, and the two chambers hold the two branches in opposite order. The undifferentiated chambers themselves give the I_{00} entry, a derivative with respect to τ_u or τ_v is the corresponding replacement times $-q_e$ by (2.8), and differentiating both ends returns the I_{11} entry because the two eigenvalues at one propagator multiply to $+1$. Only two of the four mode-function states are therefore independent, but exactly two: all four are built from the *sum* and the *difference* of the two chambers, and the two chambers are different objects, with complementary regions of time ordering and the two Hankel branches interchanged between u and v. No constant λ can relate $\mathrm{I}_{10} = \lambda\,\mathrm{I}_{00}$, since A_e and B_e coincide at the exchange-symmetric point $\Phi_e(\tau_u,\tau_v) = \Phi_e(\tau_v,\tau_u)$, where I_{10} vanishes while I_{00} does not, and are unequal at a generic point, where I_{10} does not vanish. Differentiating therefore does not return the same integral up to a phase, as it would for a block made of a *single* theta term: the theta-free massless branches of (2.18) are of that kind, which is why they collapse to one dimension and create no slot, while a full propagator keeps a genuine mode-function two-state factor. Choosing

the shared mode-function two-state $\boldsymbol{g}_e = (\mathrm{I}_{00}, \mathrm{I}_{10})^T$ and writing $\mathcal{M}_e \simeq \mathbb{C}^2$ for the two-dimensional space it spans — the single slot that the massless full propagator e contributes to the graph —, the constant embedding and projection are

$$\mathrm{I}_e^{(4)} = S_e \boldsymbol{g}_e, \qquad S_e = \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_e = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad P_e S_e = \mathbb{I}_2, \tag{2.24}$$

and every four-dimensional operator that preserves the relation subspace (2.22) reduces to $P_e O S_e$. In particular

$$P_e(\sigma_2 \otimes \mathbb{I}_2) S_e = \sigma_2, \qquad P_e(\mathbb{I}_2 \otimes \sigma_2) S_e = -\sigma_2, \tag{2.25}$$

so the two vertices joined by a massless full propagator act with opposite signs of σ_2 on one shared two-dimensional slot. Its bit r_e enters the mode-function state string of both vertices joined by e; the slot cannot be duplicated into two independent mode-function two-state factors. Consequently the full-graph tensor structure survives, but the vertex-by-vertex factorization $\mathcal{V}_s \overset{?}{=} \bigotimes_{v \in V_s} \mathcal{V}_v$ fails, because $\mathcal{M}_e$ cannot belong to both u_e and v_e at once. The reduced space of a sector is

$$\mathcal{V}_s = \bigotimes_{v \in V_s} \mathcal{E}_v \otimes \bigotimes_{i \in H(s)} \mathcal{H}_i \otimes \bigotimes_{e \in E_m(s)} \mathcal{M}_e, \qquad \dim \mathcal{V}_s = 2^{|H(s)| + |E_m(s)|}, \tag{2.26}$$

where $H(s)$ is the set of mode-function two-state factors that the massive propagators of s still contribute, one space $\mathcal{H}_i$ each, and $E_m(s)$ the set of its uncontracted massless full propagators, one shared space $\mathcal{M}_e$ each.

The theta-derivative contact vector $\boldsymbol{\rho}_e^{(4)} = (0, -1, +1, 0)^T = S_e |1\rangle_e$ fixes the shared-slot selector to $|1\rangle_e = (0, 1)^T$ by (2.24). With the propagator sign σ_e of (2.22), the vertices u_e and v_e, in that order, carry the *contact weights*

$$w_e^{(u_e)} = +2\mathrm{i}\sigma_e, \qquad w_e^{(v_e)} = -2\mathrm{i}\sigma_e, \tag{2.27}$$

in the branch pair $-\mathrm{i}e^{\mathrm{i}z}$ and $+\mathrm{i}e^{-\mathrm{i}z}$ that (2.21) leaves at $\nu = \frac{1}{2}$ once the common factor $\sqrt{2/\pi}$ is divided out; in the h mode-function state basis of (2.8) the same pair reads $\pm 4\mathrm{i}\sigma_e/\pi$. A massive propagator stays in h normalization at both ends, and its contact keeps the $4\mathrm{i}/\pi$ explicit, as in the pinch factors (2.16) and in the lowering row (2.49). In the raw h basis, the injection matrices below include the full weights $\pm 4\mathrm{i}\sigma_e/\pi$.

The package uses unit-amplitude sum and difference states. For one shared slot this is the constant change of basis

$$\widehat{\boldsymbol{g}}_e = \frac{\pi}{2} D_{\sigma_e} \boldsymbol{g}_e, \qquad D_{\sigma_e} = \mathrm{diag}(1, \mathrm{i}\sigma_e), \qquad \widehat{\boldsymbol{g}}_e = \frac{\pi}{2}(A_e + B_e,\, B_e - A_e)^{\mathsf{T}}. \tag{2.28}$$

The factor $\pi/2$ removes the two Hankel amplitudes. The transformed contact weights are $(-2, +2)$ at (u_e, v_e), while the regular generator is proportional to $\mathrm{i}\sigma_e q_e \sigma_1$ and is diago-

nalized by the Hadamard matrix $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$. On a sector, write Q_s for the tensor product of these changes on its massless shared slots and the identity on its massive slots, and set $\widehat{\mathrm{I}}_s = Q_s \mathrm{I}_s$. The transformation preserves the energy letters and is distinct from the sector normalization $\mathcal{N}_s$.

Local generators and the full-graph $\mathbf{M_1}$, $\mathbf{M_0}$. For each massless full propagator, whose two vertices we order as (u_e, v_e), define the incidence

$$\epsilon_{ve} = \begin{cases} +1, & v = u_e, \\ -1, & v = v_e, \\ 0, & v \notin e, \end{cases} \tag{2.29}$$

and let $H_v(s)$ be the set of mode-function two-state factors that the massive propagators of s contribute at v. Each building block then contributes to the vertex v the pair of atoms

$$\Delta\mathrm{M}_1^{(E_v)} = 0, \qquad \Delta\mathrm{M}_0^{(E_v)} = \mathrm{i}p_{0,v}\mathbb{I}, \qquad \text{the exponential of the vertex } v, \tag{2.30}$$

$$\Delta\mathrm{M}_1^{(h_i)} = -(1-2\nu_i)P_1, \qquad \Delta\mathrm{M}_0^{(h_i)} = -\mathrm{i}k_i\sigma_2, \qquad \text{a massive propagator of order } \nu_i \text{ at } v, \tag{2.31}$$

$$\Delta\mathrm{M}_1^{(\mathcal{M}_e)} = 0, \qquad \Delta\mathrm{M}_0^{(\mathcal{M}_e)} = -\mathrm{i}\epsilon_{ve}q_e\sigma_2, \quad \text{a massless propagator } (\nu = \tfrac{1}{2}) \text{ at } v. \tag{2.32}$$

The massive atom acts on the two-state space of a massive propagator, the massless atom on the shared slot, once per vertex with the sign ϵ_{ve}; a massless propagator does not enter M_1 at all, its $\nu = 1/2$ making $1-2\nu$ vanish identically. The complete matrices of a vertex v in a sector s are the sums of these atoms, each embedded in the slot it acts on, together with the $\nu_{0,v}\mathbb{I}_s$ of the vertex time power,

$$\begin{aligned} \mathrm{M}_{1;v,s}(\nu_{0,v}) &= \nu_{0,v}\mathbb{I}_s + \mathrm{Emb}_{\mathcal{E}_v}\Big(\Delta\mathrm{M}_1^{(E_v)}\Big) + \sum_{i\in H_v(s)} \mathrm{Emb}_i\Big(\Delta\mathrm{M}_1^{(h_i)}\Big) \\ &\quad + \sum_{e\in E_m(s)} \mathrm{Emb}_e\Big(\Delta\mathrm{M}_1^{(\mathcal{M}_e)}\Big) \\ &= \nu_{0,v}\mathbb{I}_s - \sum_{i\in H_v(s)} (1-2\nu_i)\,\mathrm{Emb}_i(P_1), \end{aligned} \tag{2.33}$$

$$\begin{aligned} \mathrm{M}_{0;v,s} &= \mathrm{Emb}_{\mathcal{E}_v}\Big(\Delta\mathrm{M}_0^{(E_v)}\Big) + \sum_{i\in H_v(s)} \mathrm{Emb}_i\Big(\Delta\mathrm{M}_0^{(h_i)}\Big) + \sum_{e\in E_m(s)} \mathrm{Emb}_e\Big(\Delta\mathrm{M}_0^{(\mathcal{M}_e)}\Big) \\ &= \mathrm{i}p_{0,v}\mathbb{I}_s + \sum_{i\in H_v(s)} \mathrm{Emb}_i(-\mathrm{i}k_i\sigma_2) + \sum_{e\in E_m(s)} \mathrm{Emb}_e(-\mathrm{i}\epsilon_{ve}q_e\sigma_2). \end{aligned} \tag{2.34}$$

The reduction only replaces the two mode-function two-state factors of a massless propagator by one shared factor: ∂_{τ_v} still acts only on the blocks belonging to v, so the vertex generators remain sums of embeddings on the common space (2.26), and different vertices

meet only on a shared massless slot, where they carry the same σ_2 and opposite ϵ_{ve}.

2.3 d log-form differential equations

Master choice and the regular block. The masters of a sector s are the integrals (2.13) at vanishing time-power shift, $\mathrm{I}_s(\mathbf{0};\boldsymbol{a})$. Collected in the bit order (2.12), they form the ordered vector

$$\mathrm{I}_s(\boldsymbol{n}) = \big(\mathrm{I}_s(\boldsymbol{n};\boldsymbol{a})\big)_{\boldsymbol{a}\in\{0,1\}^{n_s^{\mathrm{slot}}}}, \qquad \mathrm{I} = \big(\mathrm{I}_s(\mathbf{0})\big)_{\text{all sectors } s}, \tag{2.35}$$

where I is the stack of the $\mathrm{I}_s(\mathbf{0})$ over all sectors reachable by contacts. Write its connection as $\mathcal{A}$; the regular diagonal blocks admit logarithmic potentials,

$$\begin{aligned} &\mathrm{dI} = \mathcal{A}\mathrm{I}, \qquad \mathcal{A} = \sum_x \mathcal{A}_x \mathrm{d}x, \qquad \mathcal{A}_{ss} = \mathrm{d}\Omega^{\mathrm{reg}}_{ss}, \\ &\Omega^{\mathrm{reg}}_{ss} = \Omega_{\mathrm{ex},s} - \mathrm{i}\sum_{v\in V_s} U_s^{-1}\cdot\widetilde{\Omega}_{0;v,s}\cdot U_s\cdot \mathrm{M}_{1;v,s}(\nu_{0,v}+1)\,, \end{aligned} \tag{2.36}$$

where x runs over the independent kinematic variables. The full d log-form system is obtained in the normalized basis of (2.56). The derivation is that of [42]: differentiate the master vector, reduce the differentiated integrals back onto the masters by applying the one-step recurrence (2.42) iteratively, and read off the connection. The massless reduction changes nothing in that derivation except the two matrices through which the answer is written: $\mathrm{M}_{1;v,s}$ and $\mathrm{M}_{0;v,s}$ are now built on the reduced space (2.26), and the diagonalizer of [42] is replaced by the U_s of (2.40), which carries an extra factor on every shared slot. Because the reduced generators still commute slot by slot, that one U_s diagonalizes every vertex matrix at once, so (2.36) follows from the formula given in [42] by this single replacement. The two ingredients are diagonal matrices indexed by the mode-function state bits,

$$\begin{aligned} \big(\widetilde{\Omega}_{0;v,s}\big)_{\boldsymbol{ba}} &\equiv \begin{cases} -\mathrm{i}\log\kappa_{v,s;\boldsymbol{a}}, & \boldsymbol{b}=\boldsymbol{a} \\ 0, & \boldsymbol{b}\neq\boldsymbol{a} \end{cases}, \\ (\Omega_{\mathrm{ex,s}})_{\boldsymbol{ba}} &\equiv \begin{cases} -\sum_{i\in H(s)} a_i(1-2\nu_i)\log k_i, & \boldsymbol{b}=\boldsymbol{a} \\ 0, & \boldsymbol{b}\neq\boldsymbol{a} \end{cases}, \end{aligned} \tag{2.37}$$

with the energy letters

$$\kappa_{v,s;\boldsymbol{a}} = p_{0,v} + \sum_{i\in H_v(s)} (2a_i-1)k_i + \sum_{e\in E_m(s)} \epsilon_{ve}(2r_e-1)q_e, \tag{2.38}$$

where $\boldsymbol{a}$ contains both massive bits a_i and shared bits r_e. The same r_e enters the letters of u_e and v_e with opposite signs, so the letters remain linear forms in the momenta even for graphs with shared slots. Only the mode-function states of massive propagators contribute to Ω_{ex}, because a massless slot has $1-2\nu=0$.

$\mathrm{M}_{1;v,s}$ is diagonal in the original basis and all $\mathrm{M}_{0;v,s}$ are made of σ_2 (or, in the package quotient, of σ_1) on individual slots, so they commute and are simultaneously diagonalized

by one constant matrix, the Kronecker product of the massive diagonalizer T with the diagonalizer of the shared slots,

$$(\mathrm{M}_{1;v,s}[\nu_{0,v},\boldsymbol{\nu}])_{\boldsymbol{ba}} = \begin{cases} \nu_{0,v} - \sum_i a_i(1-2\nu_i), & \boldsymbol{b}=\boldsymbol{a} \\ 0, & \boldsymbol{b}\neq\boldsymbol{a} \end{cases}, \tag{2.39}$$

$$(\mathrm{T}_n)_{\boldsymbol{ba}} = \prod_{i=1}^{n} \mathrm{T}_{b_i a_i}, \qquad \mathrm{T} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -\mathrm{i} \\ -\mathrm{i} & 1 \end{pmatrix},$$

$$U_s = \bigotimes_{i\in H(s)} \mathrm{T}_i \otimes \bigotimes_{e\in E_m(s)} \mathrm{T}_e, \qquad \widetilde{\mathrm{M}}_{0;v,s} = U_s \mathrm{M}_{0;v,s} U_s^{-1} = \mathrm{diag}(\mathrm{i}\kappa_{v,s;\boldsymbol{a}}). \tag{2.40}$$

Here T diagonalizes σ_2 ($\mathrm{T}\sigma_2\mathrm{T}^{-1} = \sigma_3$); on the shared massless slots of the package quotient the same factor is instead the Hadamard matrix, which diagonalizes σ_1, and the two choices give the same set of letters (2.38) because D_{σ_e} only relabels the shared bit. The condition for one constant U_s to diagonalize every vertex matrix slot by slot is that all the Pauli directions arriving on a given slot mutually commute, which holds here because every massive slot receives only σ_2 and every shared slot only σ_2 (or, in the quotient, only σ_1). For a single vertex all of whose legs are massive the sums reduce to one term with $U = \mathrm{T}_n$ and (2.36) is the uniform formula of [43], in our convention.

No dense $2^N \times 2^N$ inversion ever appears: inverting $\mathrm{M}_{0;v,s}$ means $U_s^{-1}\,\mathrm{diag}(1/(\mathrm{i}\kappa_{v,s;\boldsymbol{a}}))U_s$, i.e. the letter-by-letter reciprocal of the scalar letters (2.40), and M_1^{-1} is the reciprocal of its diagonal entries. Changing to user or Wick coordinates does not require a new derivation of the DE; it only pulls back the one-form, and changing basis to the package quotient is the constant similarity transformation generated by the D_{σ_e}.

What the extra relations change. Three statements isolate the effect of the massless reduction on the DE. (1) A massless shared slot contributes nothing to Ω_{ex} and nothing to M_1, so the diagonal logarithmic structure in the orders ν_i is exactly the massive one. (2) It contributes one mode-function state bit r_e to the letters of both vertices with opposite incidence (2.38), so the two vertices share a bit while their recurrences remain separate and stay in closed form. (3) The theta derivative of a full propagator, massive or massless, is not a regular operator: it turns the parent sector into a sum over strictly lower sectors, which produces the off-diagonal blocks (2.59); and the base power of the merged component counts vertex merges, not contracted propagators, so for a component C it is

$$A_{C,s} = \sum_{v\in C} \nu_{0,v} + \sum_{e\in E_s(C)} \varrho_e + |C| - 1, \tag{2.41}$$

where $E_s(C)$ is the set of propagators contracted into C, and ϱ_e is the raw contact power of the propagator — $\varrho_e = 2\nu_e - 1$ for a massive full propagator and $\varrho_e = 0$ for a massless one, as will follow from (2.49) and (2.22). This ϱ_e is a fixed number of the propagator, not to be confused with the shared mode-function state bit r_e of (2.38). The last term is why a simultaneous contact of k shared-theta propagators counts as one vertex merge and not

as k (App. A).

Iterative reduction. One step of the recurrence changes the shift n_C of exactly one time component. A merged component is integrated over a single time, so the step reads the component's power and energy: below, $\nu_{0,v}$ and $p_{0,v}$ stand for $A_{C,s}$ of (2.41) and $p_{0,C}$ of (2.18) whenever v lies in a larger C, and they become the vertex quantities exactly when $C = \{v\}$; the same reading applies to (2.36)–(2.40). Writing $\boldsymbol{e}_C$ for the unit vector of the component that contains v, the complete one-step IBP at v is

$$-\mathrm{M}_{1;v,s}(A_{C,s} + n_C)\,\mathrm{I}_s(\boldsymbol{n} - \boldsymbol{e}_C) + \mathrm{M}_{0;v,s}\,\mathrm{I}_s(\boldsymbol{n}) + \mathcal{R}_{v,s}(\boldsymbol{n}) = 0, \tag{2.42}$$

where the minus in front of M_1 is the $(-\tau)^A$ derivative and $\mathcal{R}_{v,s}$ collects the theta contact terms of the propagators incident to v (next subsection). Solving (2.42) for the vector at $\boldsymbol{n} - \boldsymbol{e}_C$, and the same relation taken one step higher for the one at $\boldsymbol{n} + \boldsymbol{e}_C$, gives the two directions

$$A_{-,v,s}(\boldsymbol{n}) = \mathrm{M}_{1;v,s}(A_{C,s} + n_C)^{-1}\mathrm{M}_{0;v,s}, \tag{2.43}$$

$$A_{+,v,s}(\boldsymbol{n}) = U_s^{-1}\widetilde{\mathrm{M}}_{0;v,s}^{-1}U_s\,\mathrm{M}_{1;v,s}(A_{C,s} + n_C + 1), \tag{2.44}$$

so $\mathrm{I}_s(\boldsymbol{n} \pm \boldsymbol{e}_C)$ is expressed in terms of $\mathrm{I}_s(\boldsymbol{n})$ with only two diagonal inverses, up to $\mathcal{R}$: the lowering step carries $+\mathrm{M}_{1;v,s}^{-1}\mathcal{R}_{v,s}(\boldsymbol{n})$ and the raising step $-\mathrm{M}_{0;v,s}^{-1}\mathcal{R}_{v,s}(\boldsymbol{n} + \boldsymbol{e}_C)$. The lowering step is singular when an eigenvalue of M_1 vanishes, $A_{C,s} + n_C - \sum_i a_i(1 - 2\nu_i) = 0$; this is a property of the mode-function states and the time powers, not of the kinematics, and the regulator of Sec. 3.4 moves the solution away from this singular layer. The raising step is singular on the kinematic hypersurface $\kappa_{v,s;\boldsymbol{a}} = 0$, i.e. when the momenta become collinear and split into two opposite groups; there the DE has its poles, and the letter count is exactly the number of such hypersurfaces.

2.4 The remaining term and the contact blocks of the DE

Two different remaining terms. The lowering and raising directions evaluate the contact term $\mathcal{R}_{v,s}$ in (2.42) at different shifts. If the lower time powers of the child sector $t \prec s$ are reduced first, one obtains the *lowering* version

$$\mathcal{R}_{v,s}^{(0)} \equiv \mathcal{R}_{v,s}(\boldsymbol{n}), \qquad \text{child reached at shift } -1, \tag{2.45}$$

while the contact that appears in the raising row (2.44) is

$$\mathcal{R}_{v,s}^{(1)} \equiv \mathcal{R}_{v,s}(\boldsymbol{n} + \boldsymbol{e}_C), \qquad \text{child reached at shift } 0. \tag{2.46}$$

The reason is the base-power rule (2.41): the raw contact integrand carries one factor $(-\tau)^{\varrho_e}$ too few to be a master of the child at vanishing shift, so $\mathcal{R}^{(0)}$ lands on child shift -1 and only $\mathcal{R}^{(1)}$ — obtained by raising the parent power by one, which raises the merged component by one — hits the child master itself. Since the DE differentiates the masters

$\mathrm{I}_s(\mathbf{0})$ and therefore uses the raising row, *the off-diagonal blocks of the DE must be built from* $\mathcal{R}^{(1)}$; substituting $\mathcal{R}^{(0)}$ gives the wrong sign and a non-closing expression.

The two-vertex massive contraction. Take two vertices u, v joined by one massive full propagator with order ν at each of its two vertices, momentum magnitude k, the legs i at u and j at v; the two other legs are collected in $\boldsymbol{a}_{\hat{\imath}}$ and $\boldsymbol{b}_{\hat{\jmath}}$. The SK combination of the two h systems at u and v is, with the branch convention (2.3),

$$\begin{aligned} &h(\nu_i, a_i, -k\tau_u)\,\theta_{uv}^{(i,j)}\,h(\nu_j, b_j, -k\tau_v) \\ &\qquad \equiv h^{(1)}(\nu_i, a_i, -k\tau_u)\,\theta_{uv}\,h^{(2)}(\nu_j, b_j, -k\tau_v) + h^{(2)}(\nu_i, a_i, -k\tau_u)\,\theta_{vu}\,h^{(1)}(\nu_j, b_j, -k\tau_v), \\ &\nu_i = \nu_j \equiv \nu, \qquad k_i = k_j \equiv k, \end{aligned}$$

where the relative sign between the two terms is that of G_{++}, and G_{--} exchanges the branches. The one-step relation (2.42) differentiates the time of the vertex v at which it is taken, so what acts here is ∂_{τ_v}: besides the regular generators it produces $\partial_{\tau_v}\theta_{uv} = -\delta(\tau_u - \tau_v)$ and $\partial_{\tau_v}\theta_{vu} = +\delta(\tau_u - \tau_v)$, the delta sets the two times equal, $\tau_u = \tau_v \equiv \tau$, and the branch exchange of G_{--} reverses the difference, so one factor $-\sigma_e$ covers both contour signs and leaves the antisymmetric difference

$$\delta(\tau_u - \tau_v)\Big[h^{(1)}(\nu, a_i; z)\,h^{(2)}(\nu, b_j; z) - h^{(2)}(\nu, a_i; z)\,h^{(1)}(\nu, b_j; z)\Big], \qquad z = -k\tau. \tag{2.47}$$

If $a_i = b_j$ the bracket is zero identically, since the two summands then coincide; only complementary mode-function states survive, which is the origin of the selector $\delta_{a_i,1-b_j}$. For $a_i = 0, b_j = 1$ the bracket is the Wronskian of the two branches of the h system, and with (2.8) and $H_\nu^{(1)}\partial_z H_\nu^{(2)} - \partial_z H_\nu^{(1)} H_\nu^{(2)} = -4\mathrm{i}/(\pi z)$,

$$\begin{aligned} h^{(1)}(\nu, 0; z)\,h^{(2)}(\nu, 1; z) - h^{(1)}(\nu, 1; z)\,h^{(2)}(\nu, 0; z) &= z^{2\nu}\left(H_\nu^{(1)}\partial_z H_\nu^{(2)} - \partial_z H_\nu^{(1)} H_\nu^{(2)}\right) \\ &= -\frac{4\mathrm{i}}{\pi} z^{2\nu-1}. \end{aligned} \tag{2.48}$$

For $a_i = 1, b_j = 0$ the bracket is the negative of (2.48), so both cases are collected by the factor $(-1)^{a_i+1}$. Finally $z^{2\nu-1} = k^{2\nu-1}(-\tau)^{2\nu-1}$ on the physical branch, and the surviving $(-\tau)^{2\nu-1}$ multiplies the two explicit vertex powers, which gives the raw contact power $\varrho_e = 2\nu - 1$ quoted after (2.41). The remaining term of the lowering row is therefore

$$\mathcal{R}^{(0)}_{v;\tilde{\boldsymbol{r}},\sigma_e} = -\sigma_e\,\delta_{a_i,1-b_j}\,(-1)^{a_i+1}\,\frac{4\mathrm{i}}{\pi}\,k^{\,2\nu-1}\,\mathrm{I}_t\big(-\boldsymbol{e}_C; \boldsymbol{r}\big), \qquad \tilde{\boldsymbol{r}} = \boldsymbol{a}_i\boldsymbol{b}_{\hat{\jmath}}, \tag{2.49}$$

where v is the vertex the row is taken at, σ_e is the contour sign of the contracted propagator, of the same rule as in (2.27), k and ν are its momentum magnitude and order, t is the child sector reached by the contact, and $\mathrm{I}_t(-\boldsymbol{e}_C; \boldsymbol{r})$ is the child integral (2.13) of the merged component C, with mode-function state-bit string $\boldsymbol{r}$ packed into $\tilde{\boldsymbol{r}}$ and with the shift -1 on that single component; here $c_t = 1$ by (2.15), so $-\boldsymbol{e}_C = (-1)$. The child base power is $\nu_{0,u} + \nu_{0,v} + 2\nu - 1$, one below $A_{C,t}$ of (2.41), which is what the shift -1 records. The version needed by the DE is (2.49) with $\mathrm{I}_t(-\boldsymbol{e}_C; \boldsymbol{r}) \to \mathrm{I}_t(\mathbf{0}; \boldsymbol{r})$, whose base power is

$\nu_{0,u}+\nu_{0,v}+2\nu$ and which is an actual master of the child sector; the two-vertex contact of the first application example is written out in this notation in (4.7).

Two comments on the relation to [43] follow. First, that reference prints this equation with an extra Boltzmann factor $e^{\pi\,\mathrm{Im}\,\nu}$ and the power $k^{-2\nu-1}$: it is written in the negative-prefactor convention, which replaces $k^{2\nu-1}\to k^{-2\nu-1}$, and its basis of mode-function states at the two vertices of the contracted propagator uses conjugate Hankel orders. Our basis (2.8) uses the same order ν at both vertices, and the two bases are related by the constant identity $e^{-\pi\mu}H^{(2)}_{-\mathrm{i}\mu}=H^{(2)}_{\mathrm{i}\mu}$ for $\nu=\mathrm{i}\mu$, so that in the common-order basis the explicit Boltzmann factor is absorbed into this change of basis and does not appear in (2.49). Second, the same constant $-\sigma_e\frac{4\mathrm{i}}{\pi}k^{2\nu-1}$ is what the package assigns to the sector normalization $\mathcal{N}_t$ of the child t as its pinch factor; because the contact blocks of the normalized masters carry the ratio $\mathcal{N}_s/\mathcal{N}_t$ of (2.57), that constant cancels from the normalized DE, and the block (2.59) below has no residual pinch prefactor.

Injection structure and the contact block. For a general graph the contact part of the row at v is a sum over lower sectors,

$$\mathcal{R}^{(1)}_{v,s}(\boldsymbol{e}_C)=\sum_{t\prec s}C^{I,(v,1)}_{st}\,\mathrm{I}_t(\boldsymbol{0}), \tag{2.50}$$

where $C^{I,(v,1)}_{st}$ is the injection matrix in the raw h basis. It contains the local contact selectors, their Wronskian factors and the odd-subset coefficient of Sec. 2.6, with spectator bits preserved. It is independent of the vertex energies but can depend on the propagator momenta through factors such as $q_e^{2\nu_e-1}$. By the choice of base power, the child appears at vanishing shift; if an event leaves a nonzero shift, the recurrence is first applied inside the child.

Differentiating $e^{\mathrm{i}p_{0,C}\tau_C}$ with respect to the component energy brings down $\mathrm{i}\tau_C=-\mathrm{i}(-\tau_C)$, raising the time power by one: $\partial_{p_{0,C}}\mathrm{I}_s(\boldsymbol{n};\boldsymbol{a})=-\mathrm{i}\,\mathrm{I}_s(\boldsymbol{n}+\boldsymbol{e}_C;\boldsymbol{a})$. The raising row of (2.42) at $\boldsymbol{n}=\boldsymbol{0}$,

$$\mathrm{M}_{0;v,s}\,\mathrm{I}_s(\boldsymbol{e}_C)=\mathrm{M}_{1;v,s}(\nu_{0,v}+1)\,\mathrm{I}_s(\boldsymbol{0})-\mathcal{R}^{(1)}_{v,s}(\boldsymbol{e}_C), \tag{2.51}$$

gives

$$\partial_{p_{0,C}}\mathrm{I}_s(\boldsymbol{0})=-\mathrm{i}\,\mathrm{M}^{-1}_{0;v,s}\mathrm{M}_{1;v,s}(\nu_{0,v}+1)\,\mathrm{I}_s(\boldsymbol{0})+\mathrm{i}\,\mathrm{M}^{-1}_{0;v,s}\mathcal{R}^{(1)}_{v,s}(\boldsymbol{e}_C). \tag{2.52}$$

Since $\widetilde{\mathrm{M}}_{0;v,s}=\mathrm{diag}(\mathrm{i}\kappa_{v,s;\boldsymbol{a}})$ and $\partial_{p_{0,C}}\kappa_{v,s;\boldsymbol{a}}=\delta_{vC}$, the diagonalization in (2.40) gives exactly

$$\mathrm{M}^{-1}_{0;v,s}=U_s^{-1}\,\mathrm{diag}\!\left(\frac{1}{\mathrm{i}\kappa_{v,s;\boldsymbol{a}}}\right)U_s=\partial_{p_{0,C}}\big(U_s^{-1}\widetilde{\Omega}_{0;v,s}U_s\big)\equiv\partial_{p_{0,C}}\mathcal{K}_{v,s}, \tag{2.53}$$

with $\mathcal{K}_{v,s}\equiv U_s^{-1}\widetilde{\Omega}_{0;v,s}U_s$. Reading off the coefficient of $\mathrm{I}_t(\boldsymbol{0})$ in (2.52) gives the raw energy-contact block

$$(\mathcal{A}_{p_{0,C}})^{\mathrm{contact}}_{st}=\mathrm{i}\big(\partial_{p_{0,C}}\mathcal{K}_{v,s}\big)C^{I,(v,1)}_{st},\qquad t\prec s. \tag{2.54}$$

The injection is independent of the vertex energies, but contains momentum-dependent

Wronskian factors. Thus this energy derivative does not identify the full raw contact connection with $\mathrm{i}\,\mathrm{d}(\mathcal{K}_{v,s}C_{st}^{I,(v,1)})$.

When the sectors are ordered by the number of remaining full propagators, the raw connection $\mathcal{A} = \sum_x \mathcal{A}_x \mathrm{d}x$ is block triangular,

$$\mathrm{d}\begin{pmatrix} \mathrm{I}_{s_0} \\ \mathrm{I}_{s_1} \\ \vdots \end{pmatrix} = \begin{pmatrix} \mathcal{A}_{s_0 s_0} & \mathcal{A}_{s_0 s_1} & \cdots \\ 0 & \mathcal{A}_{s_1 s_1} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \mathrm{I}_{s_0} \\ \mathrm{I}_{s_1} \\ \vdots \end{pmatrix}, \tag{2.55}$$

so each sector is sourced only by itself and lower sectors. This permits a recursive solution starting from the lowest sectors. The package transports the complete normalized stack as one system (Sec. 3.3). Algebraic reduction proceeds downward through the hierarchy, since contacts replace a parent integral by child integrals.

The normalized system. The package combines the constant massless basis change (2.28) with the sector normalization (2.16):

$$\mathrm{J}_s = \mathcal{N}_s \widehat{\mathrm{I}}_s = \mathcal{N}_s Q_s \mathrm{I}_s, \qquad \mathrm{J} = \big(\mathcal{N}_s Q_s \mathrm{I}_s(\mathbf{0})\big)_{\text{all sectors } s}. \tag{2.56}$$

Since Q_s is constant and $\mathcal{N}_s$ depends on the momentum magnitudes, their action on the connection is

$$\begin{aligned} \mathrm{dJ} &= (\mathrm{d}\Omega^{\mathrm{J}})\mathrm{J}, \\ (\mathrm{d}\Omega^{\mathrm{J}})_{ss} &= Q_s \mathcal{A}_{ss} Q_s^{-1} + \mathrm{d}\log \mathcal{N}_s\, \mathbb{I}, \\ (\mathrm{d}\Omega^{\mathrm{J}})_{st} &= \frac{\mathcal{N}_s}{\mathcal{N}_t} Q_s \mathcal{A}_{st} Q_t^{-1}, \qquad t \prec s. \end{aligned} \tag{2.57}$$

The same transformation gives constant contact injections,

$$C_{st}^{J,(v,1)} = \frac{\mathcal{N}_s}{\mathcal{N}_t} Q_s C_{st}^{I,(v,1)} Q_t^{-1}, \qquad \widehat{\mathcal{K}}_{v,s} = Q_s \mathcal{K}_{v,s} Q_s^{-1}. \tag{2.58}$$

The normalization ratio cancels the massive Wronskian factors, while Q_s converts the massless contacts to the unit-amplitude basis. The normalized contact potential is therefore

$$(\Omega_{st}^{\mathrm{J}})^{\text{contact}} = \mathrm{i} \sum_{v \in V_s} \widehat{\mathcal{K}}_{v,s} C_{st}^{J,(v,1)}, \qquad t \prec s. \tag{2.59}$$

On the diagonal, the potential is $Q_s \Omega_{ss}^{\text{reg}} Q_s^{-1} + \log \mathcal{N}_s\, \mathbb{I}$. Here $\mathrm{d}\log \mathcal{N}_s = \sum_e (2\nu_e - 1)\mathrm{d}\log q_e$, with the sum over the contracted massive propagators of (2.16), so this term also has d log form.

To relate (2.59) to the implementation, denote the matrices returned by the contact maps by $C_{st}^{\text{pkg},(v,1)}$. Their massive atomic selector is the negative of the normalized physical selector, whereas their massless atom has the same weights $(-2, +2)$. For an event selecting

an odd subset E_{st}, define

$$\chi(E_{st}) = (-1)^{N_0(E_{st})}, \qquad C_{st}^{J,(v,1)} = -\chi(E_{st}) C_{st}^{\mathrm{pkg},(v,1)}, \tag{2.60}$$

where N_0 counts the selected massless full propagators. Indeed, the number of selected massive propagators is $|E_{st}| - N_0$, whose parity gives $(-1)^{|E_{st}|-N_0} = -\chi$. The component logarithmic kernel in the implementation is $-\mathrm{i}\widehat{\mathcal{K}}_{v,s}$, so its contact potential is $-\mathrm{i}\sum_v \chi(E_{st})\widehat{\mathcal{K}}_{v,s} C_{st}^{\mathrm{pkg},(v,1)}$, equal to (2.59). The vertex phase sign enters through the signed energies and their chain rule, with no additional contour factor in the injection.

2.5 Boundary conditions by nested blow-up

The last ingredient is the boundary condition in the large-energy limit. Two mechanisms make it computable: the damping exponential localizes each time integral near $\tau_v = 0$, where every factor except the thetas reduces to a power of $-\tau$, so each single-time integral becomes a Gamma function; and the strict energy rank fixes every theta to 0 or 1, so no theta couples two times and every master factorizes into a product of single-time integrals. We define the *damping energies* by $K_v \equiv \mathrm{i}p_{0,v} = \mathrm{i}\varsigma_v k_{0,v}$ and take $k_{0,v} = -\mathrm{i}\varsigma_v K_v$ with $K_v > 0$. On these damping rays, the exponential of (2.17) becomes $e^{K_v\tau_v}$, and a merged component has $K_C = \mathrm{i}p_{0,C}$. The package records the substitution as `dampingDefinition` in the vertex coordinates. The energy hierarchy below is understood on these rays, ordered by the positive damping energies. The limit $K_v \to \infty$ is one multivariate limit in which all energies diverge together, and the point it approaches is in general a degenerate multivariate singularity of the differential equation: several singular divisors meet there, so a direct series expansion about that point is not allowed. Blowing the point up first makes the expansion well defined; the nested coordinates introduced below are exactly the blow-up charts that replace the degenerate point by a chain of one-energy limits. This continues the procedure of [43], where the boundary condition of the multivariate hypergeometric system is fixed by the same resolution of its degenerate point. In a c_V-vertex correlator there are c_V exponential factors with energies $k_{0,v}$; assuming the ordering

$$k_{\sigma(1)} \gg k_{\sigma(2)} \gg \cdots \gg k_{\sigma(c_V)} \gg 1, \tag{2.61}$$

with a permutation σ, we define the nested coordinates

$$x_j = \frac{k_{\sigma(j+1)}}{k_{\sigma(j)}} \quad (1 \le j < c_V), \qquad x_{c_V} = \frac{1}{k_{\sigma(c_V)}}, \tag{2.62}$$

so that

$$\frac{1}{k_{\sigma(j)}} = \prod_{r=j}^{c_V} x_r, \qquad x_1, \ldots, x_{c_V} \longrightarrow 0. \tag{2.63}$$

The exponential localization scales satisfy

$$|\tau_{\sigma(1)}| \ll |\tau_{\sigma(2)}| \ll \cdots \ll |\tau_{\sigma(c_V)}| \ll 1. \tag{2.64}$$

Since all τ approach zero from the negative side, large k corresponds to small $|\tau|$; hence $k_u \gg k_v$ implies $\tau_u > \tau_v$, and therefore

$$\theta(\tau_u - \tau_v) \longrightarrow 1, \qquad \theta(\tau_v - \tau_u) \longrightarrow 0. \tag{2.65}$$

For G_{++} or G_{--}, (2.65) applies to every explicit theta in the propagator expansion according to the order of its argument. For a bundle of propagators that share one theta (the *common-theta bundle* of Sec. 2.6) it applies to that single theta, which is the boundary counterpart of Sec. 2.6: the bundle contributes one frozen 0 or 1, not one per propagator. If a term requires a cyclic inequality, for example

$$\theta(\tau_1 - \tau_2)\theta(\tau_2 - \tau_3)\theta(\tau_3 - \tau_1), \tag{2.66}$$

then it is automatically zero in every strict-total-order chart, instead of producing a new boundary region.

After the thetas are frozen, every master integral — in the top sector as well as in every subsector — factorizes into one one-dimensional time integral per *time component* of the sector: a component is integrated over its single time, and the legs it carries enter that integral through their mode-function small-τ coefficients. A massive leg behaves like (2.5) and a massless leg contributes its leading power or a pure exponential, so each single-time integral has the form $\int_{-\infty}^{0} \mathrm{d}\tau\, (-\tau)^A e^{\mathrm{i}K\tau} = (-\mathrm{i})^{A+1}\Gamma(A+1)K^{-A-1}$, which gives a Gamma function; worked examples of the resulting boundary data and of the blow-up charts are given in [42, 43]. The boundary conditions of all master integrals are therefore products of Gamma functions with some overall coefficients, as used in the numerical evaluation of Sec. 3.3.

Each h factor is expanded at small τ and integrated term by term. In the convention (2.8) the mode-function two-state block behaves as

$$\begin{aligned} h^{(1)}(\nu, 0; z) &= c_2\big(1 + O(z^2)\big) + c_1 z^{2\nu}\big(1 + O(z^2)\big), \\ h^{(1)}(\nu, 1; z) &= O(z) + 2\nu\, c_1\, z^{2\nu-1}\big(1 + O(z^2)\big), \end{aligned} \tag{2.67}$$

with $c_{1,2}$ as in (2.5) and $z = -k\tau$; the second Hankel branch has the same two powers with its own constants. For a vertex with a single massive leg of order ν_1, the master integral of the undifferentiated state is

$$\begin{aligned} \int_{-\infty}^{0} \mathrm{d}\tau\, (-\tau)^{\nu_0} e^{\mathrm{i}K\tau}\, h^{(1)}(\nu_1, 0; -k\tau) &= c_2\, (-\mathrm{i})^{\nu_0+1}\Gamma(\nu_0+1)K^{-\nu_0-1} \\ &\quad + c_1\, k^{2\nu_1}(-\mathrm{i})^{\nu_0+2\nu_1+1}\Gamma(\nu_0+2\nu_1+1)K^{-\nu_0-2\nu_1-1} + \cdots, \end{aligned} \tag{2.68}$$

where each term of the small-τ expansion was integrated with the one-dimensional formula above, $\int_{-\infty}^{0} \mathrm{d}\tau\, (-\tau)^A e^{\mathrm{i}K\tau} = (-\mathrm{i})^{A+1}\Gamma(A+1)K^{-A-1}$. For generic nonresonant orders the constant term anchors the undifferentiated state and the $z^{2\nu-1}$ term the differentiated state; the differentiated master integral is treated identically and anchors at the exponent $\nu_0 + 2\nu_1$. The corrections sit one integer power above the anchor of the other state, the

Frobenius separation described after eq. (3.16) of [43]. The two masters of such a vertex therefore anchor at the leading inverse-energy exponents $\nu_0 + 1$ and $\nu_0 + 2\nu_1$. Products of several h factors are handled the same way: the product is expanded in small τ, and each term integrates to one Gamma-function coefficient. Beyond these leading coefficients the local series is generated by the DE, not by the boundary data.

The construction requires a damping energy at every vertex to rank it in (2.64) and localize its time integral. A vertex without external legs has no exponential factor, so the strict-order chart is undefined there. The package introduces an *auxiliary* external-leg energy $k_{0,v}^{\mathrm{aux}}$ in the same d log-form system. It enters the energy letters and component signed energies as a physical $k_{0,v}$ does. Boundary data are constructed at $K_v^{\mathrm{aux}} \to \infty$, then transported to the physical target $k_{0,v}^{\mathrm{aux}} = 0$. All auxiliary coordinates enter the system and every requested point together, giving a single transport in the enlarged coordinate space. The package handles these coordinates internally. The zero target can be singular: if a component's only nonzero signed energy is auxiliary, its letters vanish at $k_{0,v}^{\mathrm{aux}} = 0$. At such a finite singular point, the package returns a finite value or reports an explicit divergence.

2.6 Several propagators sharing one theta

Up to now every contact has been attached to one full propagator. Two or more full propagators can, however, join the same pair of current components, so that all of them carry the same theta argument $\Delta = \tau_u - \tau_v$; the package groups exactly such propagators into a *common-theta bundle* $\mathcal{B}$, with $|\mathcal{B}| \geq 2$. This happens in a time graph that carries a cycle of time integrations at fixed internal momenta, the class read by `MSInitTimeGraph` of Sec. 3.2, which the same construction covers. A strict tree never reaches this case: contracting propagators of a tree leaves a tree, hence never two propagators on one and the same pair of components. Either the bundle is part of the graph from the start, or it appears at an intermediate stage, when a first contact merges two vertices of a cycle and leaves the remaining propagators of that cycle on one common theta. This case is absent from [42], where each contact contracts a single propagator. Figure 1 shows the minimal graph of the first kind: two vertices, no external legs, and four full propagators that all join the same pair of vertices. The handling is best summarized in four steps, with the details in App. A.

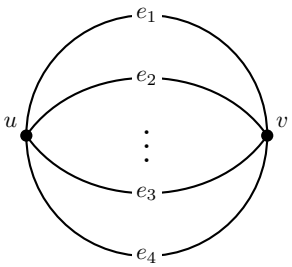


Figure 1. A common-theta bundle of four full propagators. The vertices u and v carry no external legs, and every propagator $e_1, \dots, e_4$ joins this same pair of vertices, hence the same pair of current components at every stage of the reduction. All four propagators therefore carry the identical theta argument $\Delta = \tau_u - \tau_v$ and form one bundle $\mathcal{B}$ with $|\mathcal{B}| = 4$. The two vertices sit at the left and the right end, the outer two propagators make up the halves of one circle, and the vertical dots between the inner two propagators stand for any number of further propagators joining the same pair, so the bundle is not restricted to the four propagators drawn.

For each propagator e of the bundle, write α_e for the coefficient of $\theta(\Delta)$ and β_e for the coefficient of $\theta(-\Delta)$.

1. *Multiply the thetas out first.* Since inside a bundle all thetas are the same function, $\theta(\Delta)^2 = \theta(\Delta)$ and $\theta(\Delta)\theta(-\Delta) = 0$, the product of $|\mathcal{B}|$ propagators collapses to *two* terms rather than $2^{|\mathcal{B}|}$,

$$\prod_{e\in\mathcal{B}} \Big(\theta(\Delta)\alpha_e + \theta(-\Delta)\beta_e\Big) = \theta(\Delta)\prod_{e\in\mathcal{B}} \alpha_e + \theta(-\Delta)\prod_{e\in\mathcal{B}} \beta_e. \tag{2.69}$$

2. *One theta, one delta.* Differentiating the shared theta — and only the theta, the two coefficients being left as spectators — yields a single $\delta(\Delta)$ whose coefficient is the antisymmetric difference under $\alpha \leftrightarrow \beta$, $\big(\partial_\Delta\theta(\Delta)\big)\prod_{e\in\mathcal{B}}\alpha_e + \big(\partial_\Delta\theta(-\Delta)\big)\prod_{e\in\mathcal{B}}\beta_e = \delta(\Delta)\big(\prod_{e\in\mathcal{B}}\alpha_e - \prod_{e\in\mathcal{B}}\beta_e\big)$, so all contact information of the bundle is concentrated in one distribution and the two vertices are forced to coincide only once. The terms in which the derivative acts on α_e or β_e instead are not contact terms: they are the regular part of the derivative, already carried by the M_0 and M_1 generators of Sec. 2.2, and they do not merge any component (App. A).

3. *Split and change basis.* Telescoping the antisymmetric difference and then collecting in $\mathsf{d}_e = \alpha_e - \beta_e$ and $\mathsf{m}_e = (\alpha_e + \beta_e)/2$ leaves only terms with an odd number of d factors, each with a numerical coefficient,

$$\prod_{e\in\mathcal{B}} \alpha_e - \prod_{e\in\mathcal{B}} \beta_e = \sum_{\substack{\varnothing\neq S\subseteq\mathcal{B}\\ |S| \text{ odd}}} 2^{1-|S|} \prod_{e\in S} \mathsf{d}_e \prod_{e\notin S} \mathsf{m}_e. \tag{2.70}$$

4. *Read off the events.* Every non-empty odd subset S is one simultaneous contact event: the propagators of S are contracted at once, their two vertices merge once,

the component count drops by one, and the base power of the merged component follows (2.41) with $|C|-1$ merge terms. The injection matrix $C^{I,(v,1)}_{st}$ of (2.50) for such an event is the tensor product of the local selectors and Wronskian factors of the propagators in S, multiplied by $2^{1-|S|}$ and by the spectator factors m_e.

The package uses the collected form (2.70) rather than the telescoping sum term by term, because enumerating the terms one by one would count the same sector transition several times; the argument, the parity proof and the treatment of unselected propagators whose two vertices become coincident are given in App. A. The same symmetric part governs a propagator whose two vertices have become coincident, its boundary data included: the two branch arrangements are replaced by their mean, one mean per propagator.

3 Using the package

The package turns the formulas of Sec. 2 into a sequential workflow: initialize a family from an input specification, obtain its masters, reduction system and d log-form DE, then transport along a kinematic path with the bundled backend and, where the time integrals diverge, reconstruct the Laurent series in the analytic regulator. This section describes installation (Sec. 3.1), the input formats and the functions of each stage (Sec. 3.2), the numerical interface with its bundled backend (Sec. 3.3), the Laurent reconstruction (Sec. 3.4) and the options with validation and common pitfalls (Sec. 3.5). Sec. 4 then exercises the workflow on the distributed examples.

3.1 Installation and loading

System requirements.

- A Wolfram Language kernel: Mathematica or the free Wolfram Engine (the package is a `.wl` package; no external Wolfram libraries are needed).
- Python 3.10 or newer with `python-flint>=0.6` and `sympy>=1.12` installed, for the numerical backend. `MadStree` follows a zero-probe policy: the interpreter is selected only by the explicit `PythonExecutable` option or, when that is `Automatic` (the default), by the `MADSTREE_PYTHON` environment variable, falling back to `"python"`.

Install the dependencies in the Python interpreter used for numerical evaluation:

```
python -m pip install "python-flint>=0.6" "sympy>=1.12"
```

Here `python` should be replaced by the selected interpreter if necessary. No further backend installation is needed: `MadStree` bundles `FlintNDE`, a solver for first-order matrix differential equations built on the exact ball arithmetic of `python-flint`, under `MadStree/Vendor/FlintNDE`, and delegates every numerical task of this package to this bundled copy. Its role in the numerical workflow is described in Sec. 3.3.

Loading the package. Download or clone the official release repository https://github.com/J7C/dSIBP-Packages and set `madStreeDirectory` to the `MadStree` directory in the local repository, add it to `$Path`, and load with `Needs`:

```
madStreeDirectory = ".../dSIBP-Packages/MadStree";
AppendTo[$Path, madStreeDirectory];
Needs["MadStree`"];
```

Replace the example path with the absolute path to the local `MadStree` directory. This loads `MadStree.m`, which loads `Kernel/MadStree.wl` with explicit UTF-8. Equivalently, load the kernel file directly:

```
Get[FileNameJoin[{madStreeDirectory, "Kernel", "MadStree.wl"}],
  CharacterEncoding -> "UTF-8"];
```

Loading through `Needs` requires no user encoding option. `MadStree` uses its bundled `FlintNDE` backend without requiring `Needs["FlintNDE`"]`. Use a fresh kernel when switching versions, since `Needs` does not reload an already loaded package.

After loading, `MSFlintNDEConfiguration[]` returns an association whose `"version"` entry should be `"1.0"` and whose `"availableQ"` entry should be `True`. The second value confirms that `flintnde/__init__.py` exists under the configured backend directory, which is `Vendor/FlintNDE` relative to `madStreeDirectory` by default.

```
MSFlintNDEConfiguration[]["version"]
```

```
MSFlintNDEConfiguration[]["availableQ"]
```

3.2 Input formats and core functions

An initialized context is the root of the whole interface: the input specification fixes the family, and the masters, the reduction system and the d log-form DE are derived from it rather than re-specified. We first describe the three initialization formats — trees, loop time integrals and single-vertex families — together with the package conventions, then the core functions and the organization of their outputs.

Tree diagrams: `MSInitTree`. A tree is given by an association of vertices and propagators, with propagators listed under the input key `"lines"`. A tree with two vertices and one massless propagator between them:

```
spec = <|
  "name" -> "masslessFullEdge",
  "vertices" -> {
    <|"id" -> v1, "externalLegEnergy" -> k1,
      "timePower" -> a1, "vertexType" -> "+"|>,
    <|"id" -> v2, "externalLegEnergy" -> k2,
      "timePower" -> a2, "vertexType" -> "+"|>
  },
```

```
  "lines" -> {
    <|"type" -> "massless", "endpoints" -> {v1, v2},
      "momentum" -> q, "nu" -> 1/2|>
  }
|>;
context = MSInitTree[spec];
```

Tables 3.1 and 3.2 summarize the graph input fields.

field	meaning	note
`id`	vertex symbol	required
`externalLegEnergy`	external energy parameter (e.g. `k1`)	optional; default 0
`timePower`	time-power variable a_v (e.g. `a1`)	required
`vertexType`	contour assignment `"+"` or `"-"`	required

Table 3.1. Vertex fields for tree and time-graph input.

field	meaning	note
`type`	`"massive"` or `"massless"`	required
`endpoints`	list of one external or two internal vertices	required
`momentum`	momentum variable k_i (e.g. `q`)	required
`nu`	supplied Bessel/Hankel order ν	required for massive propagators; massless default 1/2

Table 3.2. Propagator fields for tree and time-graph input.

A massless propagator joining two vertices with equal contour signs is classified internally as `masslessFull` and uses the shared two-state quotient basis of Sec. 2.2. For opposite contour signs it is theta-free and contributes no state slot. The `"massless"` input is the pure exponential block at the fixed massless order `nu -> 1/2` of Sec. 2.1; propagators of other half-integer orders must first be expanded as described there. There is no input field selecting an uncompressed massless basis. The initialization option `NuConvention` takes `"Positive"` (default) or `"Negative"` and chooses the prefactor sign of $h(\nu, 0; z) = z^{\pm\nu} H_\nu(z)$ introduced in Sec. 2.1; it is fixed for the whole context and cannot be changed per call.

Propagator numbers are assigned from the input order. The Full, Cross or External class and the contour signs are derived from the `vertexType` of the two vertices the propagator joins; they are not supplied as a type value or as separate sign fields. A `"+"` vertex has phase $e^{-\mathrm{i}E\tau}$ and a `"-"` vertex has phase $e^{+\mathrm{i}E\tau}$, where E is `externalLegEnergy`. An omitted or zero external energy is represented internally by an auxiliary variable whose target value is zero. Full propagators that join the same pair of vertices are grouped into

one common-theta bundle automatically; the optional top-level field `"thetaBundles"` pins such a group explicitly as a list of propagator ids, and any group whose members do not join the same pair of vertices is rejected.

Loop diagrams without loop integral: `MSInitTimeGraph`. For a pure time-order graph the vertex and propagator fields are the same, but the graph may contain cycles. A triangle cycle reads

```
cycleContext = MSInitTimeGraph[<|
  "vertices" -> {
    <|"id" -> t1, "externalLegEnergy" -> k1,
      "timePower" -> a1, "vertexType" -> "+"|>,
    <|"id" -> t2, "externalLegEnergy" -> k2,
      "timePower" -> a2, "vertexType" -> "+"|>,
    <|"id" -> t3, "externalLegEnergy" -> k3,
      "timePower" -> a3, "vertexType" -> "+"|>
  },
  "lines" -> {
    <|"type" -> "massless", "endpoints" -> {t1, t2},
      "momentum" -> q12, "nu" -> 1/2|>,
    <|"type" -> "massless", "endpoints" -> {t2, t3},
      "momentum" -> q23, "nu" -> 1/2|>,
    <|"type" -> "massless", "endpoints" -> {t3, t1},
      "momentum" -> q31, "nu" -> 1/2|>
  }
|>];
```

Here all three propagators join vertices with equal contour signs, so they are classified as `masslessFull`. Once one of them is contracted, the remaining two connect the same pair of components and therefore form the common-theta bundle of Sec. 2.6.

Single-vertex integral families: `MSInitVertexFamily`. A single vertex needs no graph input. The compact input follows the reference `ki/nui` convention,

```
vertexContext = MSInitVertexFamily[<|
  "ki" -> {k0, k1, k2},
  "nui" -> {a0, nu1, nu2},
  "hankelBranches" -> {1, 2}
|>, NuConvention -> "Positive"];
```

where `First[ki]` is the pure vertex phase energy and `First[nui]` the base power of $(-\tau)$; the remaining entries give, position by position, the h-block momenta and their Hankel orders ν, which are used exactly as in Sec. 2.1. The explicit alternative `externalLegEnergy/timePower/hBlocks/exponentialBlocks` is also accepted, where

each h-block is an association with the keys `momentum`, `nu` and `hankelBranch`. In contrast to the tree and time-graph functions, `MSInitVertexFamily` rejects any field outside its own key set.

Package conventions. The package implements the h mode-function states of (2.8) with $z = -k\tau$ and the single function name h, using

$$h(\nu, 0; z) = z^{\pm\nu} H_\nu(z), \tag{3.1}$$

where the plus sign is the default (`NuConvention->"Positive"`) and the minus sign gives the convention of [42]; the two bases are related by $\nu \to -\nu$ (Sec. 2.1). The master integrals are the $\mathrm{I}_s(\mathbf{0}; \boldsymbol{a})$ of (2.35), which on a single vertex is the one-vertex family $\mathrm{I}^{(0)}_{\boldsymbol{a};s_1\cdots s_n}$ of (2.11); for multi-vertex correlators they are the products and sub-sector integrals of Sec. 2.3. The package represents their normalized counterparts $J_s = \mathcal{N}_s \widehat{\mathrm{I}}_s$ by `MSIntegral[sectorKey, aShifts, stateBits]`: `sectorKey` is a binary string over the root propagators in the input order, with `"1"` for an uncontracted propagator and `"0"` for a propagator contracted by a contact; `aShifts` assigns one integer shift of the base time power to every vertex component, where a component is a maximal set of vertices merged by the contractions; `stateBits` gives the mode-function state of every two-state slot of the sector, in slot order; the slot order itself is that of `MSSlotRegistry[context, sector]`. The bare counterpart $\mathrm{I}_s(\boldsymbol{n}; \boldsymbol{a})$ with the same three indices is available as the inert `MSBareIntegral[sectorKey, aShifts, stateBits]` for display and audit, and `MSIntegralDefinition[integral, context]` returns for any single valid `MSIntegral` its bare label, the exact sector normalization $\mathcal{N}_s$ of (2.16) — one pinch factor $-\sigma_e \frac{4\mathrm{i}}{\pi} q_e^{2\nu_e - 1}$ per contracted massive full propagator — and the inert identity $J_s(\boldsymbol{n}; \boldsymbol{a}) = \mathcal{N}_s \, \mathrm{I}_s(\boldsymbol{n}; \boldsymbol{a})$. Their connection is the normalized d log-form system (2.57).

We give a few examples of `MSIntegral[sectorKey, aShifts, stateBits]`. For the three-vertex chain v_1–v_2–v_3 with internal propagators l_{12}, l_{23}, vertex energies k_1, k_2, k_3 and time powers a_1, a_2, a_3, the contact-reachable sectors are the four strings `"11"`, `"01"`, `"10"`, `"00"`. Table 3.3 shows their master indices for three choices of propagators.

(i) both propagators massive: l_{12} with ν_{12}, l_{23} with ν_{23}.

`sectorKey`	contracted	components	master integrals
`"11"`	—	$\{v_1\},\{v_2\},\{v_3\}$	`MSIntegral["11",{0,0,0},{a,b,c,d}]` (16)
`"01"`	l_{12}	$\{v_1,v_2\},\{v_3\}$	`MSIntegral["01",{0,0},{c,d}]` (4)
`"10"`	l_{23}	$\{v_1\},\{v_2,v_3\}$	`MSIntegral["10",{0,0},{a,b}]` (4)
`"00"`	l_{12},l_{23}	$\{v_1,v_2,v_3\}$	`MSIntegral["00",{0},{}]` (1)

(ii) l_{12} massive with ν_{12}, l_{23} massless.

`sectorKey`	contracted	components	master integrals
`"11"`	—	$\{v_1\},\{v_2\},\{v_3\}$	`MSIntegral["11",{0,0,0},{a,b,r}]` (8)
`"01"`	l_{12}	$\{v_1,v_2\},\{v_3\}$	`MSIntegral["01",{0,0},{r}]` (2)
`"10"`	l_{23}	$\{v_1\},\{v_2,v_3\}$	`MSIntegral["10",{0,0},{a,b}]` (4)
`"00"`	l_{12},l_{23}	$\{v_1,v_2,v_3\}$	`MSIntegral["00",{0},{}]` (1)

(iii) both propagators massless.

`sectorKey`	contracted	components	master integrals
`"11"`	—	$\{v_1\},\{v_2\},\{v_3\}$	`MSIntegral["11",{0,0,0},{r,s}]` (4)
`"01"`	l_{12}	$\{v_1,v_2\},\{v_3\}$	`MSIntegral["01",{0,0},{r}]` (2)
`"10"`	l_{23}	$\{v_1\},\{v_2,v_3\}$	`MSIntegral["10",{0,0},{r}]` (2)
`"00"`	l_{12},l_{23}	$\{v_1,v_2,v_3\}$	`MSIntegral["00",{0},{}]` (1)

Table 3.3. Master indices of an equal-sign three-vertex chain with massive or massless propagators. Each state index takes the values 0 and 1; parentheses give the number of masters in each sector.

The vertex `vertexType` field (`"+"` or `"-"`) determines the sign of the pure vertex exponential $e^{\mathrm{i}k_0\tau}$ of (2.11), and the pair of vertex types determines the SK class of each propagator, from which the default Hankel branches of (2.8) follow; a propagator may override them with its own `hankelBranches` field. The package h mode-function states are those of (2.8), i.e. $h(\nu,1;z) = \partial_z h(\nu,0;z)$. `MSHTohMatrix[nu, z, context]` and `MShToHMatrix[nu, z, context]` give the local 2×2 change of basis between the raw Hankel states $(H_\nu(z),\partial_z H_\nu(z))$ and these states and its inverse; `MSConvertBasis` applies it to a local two-component state vector or, given a sector key and component times, to the complete sector state vector as the Kronecker product over the slot registry, with the massless shared mode-function two-state slot of Sec. 2.2 acting trivially. These commands convert integrand states, not integral objects: the powers of $z=-k\tau$ in the matrix also shift the component base time powers, so converting an `MSIntegral` itself fails closed.

Once the context is initialized, the following functions expose the formula products of Sec. 2.

Core functions. Most functions take the initialized `context`; `MSExportEvaluationData` takes a completed evaluation result instead. Table 3.4 lists the principal functions.

function	purpose
`MSSectors[context]`	contact-reachable sectors in deterministic order (keys, contracted propagators, master counts)
`MSMasterIntegrals[context]`	ordered master records; each `"integral"` field is an `MSIntegral[sectorKey, aShifts, stateBits]`
`MSIntegralDefinition[integral, context]`	bare label, exact normalization $\mathcal{N}_s$ and inert identity of one valid `MSIntegral`
`MSHTohMatrix[...]` / `MShToHMatrix[...]` / `MSConvertBasis[...]`	local and sector-level conversion between raw Hankel states and h states
`MSFormulaMatrices[context, key]`	per-component M_1, M_0 and energy letters $\kappa_{\boldsymbol{a}}$; sector diagonalizer `"U"` and `"UInverse"`
`MSDLogDE[context]`	block-triangular d log-form potential Ω, letters, certification status
`MSRecurrenceStep[integral, component, context]`	one recurrence step moving one nonzero time shift toward zero
`MSReduce[expr, context]`	iterative IBP reduction to the master basis; scalar, list or nested table input
`MSContactMaps[context, key]`	parent-to-sub-sector contact maps
`MSWriteFormulaArtifacts[context]`	write formula products beside the caller
`MSFormulaData[context]`	aggregate all-sector masters, recurrence metadata and the d log-form DE
`MSEvaluatePath[context, pointSequence, opts]`	single-stage evaluation: boundary initialization and segment transport in one backend process; optional FlintNDE path planning
`MSReconstructEpSeries[context, ep, pointSequence, opts]`	Laurent series in the common analytic regulator: certified lowest power, automatic sample points, incremental fitting
`MSExportEvaluationData[evaluation, opts]`	export the saved ordinary-point records of an evaluation to CSV/JSON

Table 3.4. Principal formula, evaluation and export functions.

`MSSectors[context]` returns the sector list. Each sector key is the fixed-length bit string over the root propagators in input order (`"0"` contracted, `"1"` uncontracted); the sub-sector structure is discussed in Sec. 2.3. Keys are strings, not integers; leading zeros are part of the identity.

`MSMasterIntegrals[context]` returns the ordered master records. Their `"integral"` fields contain the `MSIntegral[sectorKey, aShifts, stateBits]` objects of Sec. 3.2, i.e. the normalized masters $J_s(\mathbf{0};\boldsymbol{a})$ of (2.56). Use `Lookup[MSMasterIntegrals[context], "integral"]` to obtain this ordered basis. Each record also contains `"bareIntegral"`, `"normalization"` and `"definition"`.

All matrices, boundary vectors and numerical results follow this order.

`MSFormulaMatrices[context, key]` returns the component records under `"components"`, with fields `"M1"`, `"M0"` and `"energyLetters"` for the matrices and letters of Sec. 2.2. The sector diagonalizer of M_0 and its inverse are returned as `"U"` and `"UInverse"`; M_1 is already diagonal in the mode-function state basis. Here `key` is a sector-key string; omitting it, or passing `All`, returns an association for all sectors. `MSContactMaps` accepts the same sector selection.

`MSDLogDE[context]` returns `"omegaPotential"`, the matrix Ω of Sec. 2.3 whose derivative along a path is the DE matrix, together with the letters (the arguments of the logarithms of (2.37)), `"letterMatrices"`, `"dlogResidual"`, `"constantResidueQ"` and the certification `"dlogStatus"`. The closure test itself reduces the shifted integrals generated by the connection with `MSReduce`, requiring status `"reduced"`, a zero `"nonMasterResidual"` and no `"remainingShiftedIntegrals"` on every entry; the connection matrix is then read off the reduction coefficient vectors. Only the status `"certifiedByFormulaChecks"` certifies a closed d log-form, and only a certified potential may be used numerically.

`MSReduce[expr, context, MasterBasis -> Automatic]` reduces finite linear combinations of legal `MSIntegrals`, or lists and nested tables of such expressions, by the lowering/raising recurrences $A_\pm$ of Sec. 2.3 along the sub-sector structure of Sec. 2.3. The single step is exposed separately as `MSRecurrenceStep[integral, component, context]`, which moves one nonzero time shift of one component toward zero; with the component omitted it picks the first nonzero shift, and it refuses any step that would move a component into one of the divergent small-time layers fixed at initialization. Results of `MSReduce` include `"result"`, `"masterBasis"`, `"coefficientVector"`, `"masterRules"`, `"nonMasterResidual"`, `"remainingShiftedIntegrals"` and `"singularLayers"`; a full reduction requires a zero residual and an empty shifted list for every scalar expression. For list input, these fields preserve the input shape; `"elementResults"`, `"failurePositions"` and `"incompletePositions"` provide the element-level reports. The reduction runs on the time powers fixed at initialization: a context initialized with specialized or numerical `timePower` values, such as the regulated powers of Example 06, reduces with the substitution in place, and purely symbolic results may equivalently be substituted afterwards. A worked reduction of a two-term shifted combination is displayed in Sec. 4.2.

`MSFormulaData[context]` returns all-sector masters, the per-sector $M_1/M_0/T_n$ recurrence metadata of Sec. 2.2 and the complete block-triangular d log-form DE in the master order of `MSDLogDE`; `TimePowerRules` may substitute the vertex time powers a_i in the recurrence metadata, but the returned `"dlogDE"` remains analytic and unsubstituted.

`MSEvaluatePath[context, pointSequence, opts]` is the single-stage numerical entry for evaluating the master integrals at the user points. `pointSequence` is a coor-

dinate table in the format of Sec. 3.3. Fixed parameters are supplied once through `ParameterRules`. Boundary generation, transport and result records are described in Sec. 3.3; the principal options and point tags are summarized in Sec. 3.5.

`MSReconstructEpSeries[context, ep, pointSequence, opts]` reconstructs the Laurent series of the master integrals in a common analytic regulator `ep`, with `MaximumEpPower` the highest returned power (default 0, i.e. the pole terms and the finite part). It uses the same `pointSequence` and `ParameterRules` format as `MSEvaluatePath`. Only `ParameterRules`, not the coordinates, may depend on `ep`. By default, no `ep` values need to be supplied; `EpSamplePoints` and `EpValidationPoints` may instead specify the production points and disjoint validation points. The certificate conditions, sampling and fitting procedure are described in Sec. 3.4; the corresponding options are summarized in Sec. 3.5, and a complete application is presented in Sec. 4.2.

`MSExportEvaluationData[evaluation, opts]` exports the saved ordinary-point records of one `MSEvaluatePath` result, including the coordinates, the real/imaginary parts of every master, `"status"`, `"relativeDifferenceInf"` and `"targetRelativeErrorMet"`, to CSV and JSON for downstream plotting tools; transient records are excluded. The two precision fields are the corresponding segment reports, not separate error estimates computed at every point. `ExportFormats` selects CSV, JSON or both, and `SignificantDigits` defaults to 16; see Sec. 3.2.

Output organization.

- Transient request JSON and backend logs: `results_temp/nde/` beside the caller by default. Request and log files are deleted after a successful run and retained for backend failures; the package also deletes them for the structured `"singularPathRefused"` and `"leadingOrderRefused"` outcomes, which are results rather than failures. Successful JSON results remain in `results_temp/cache/` for reuse. `MSRuntimeDirectory` may specify a different runtime directory.

- Saved point records: `MSEvaluatePath` returns the records under `"pointResults"`. An ordinary untagged coordinate row has `"status" -> "saved"`; `"saved"` is a status value, not a top-level key. A row `{{values...}, "tmp"}` gives a transient waypoint, represented by `"status" -> "transient"` and excluded from export. Singular-point classifications are returned under `"singularityClassifications"` by the same backend evaluation run.

- Formula artifacts: initialization itself writes no files. `MSWriteFormulaArtifacts[context, opts]` — with its own options `TimePowerRules`, which specializes the recurrence metadata only, and `MSOutputDirectory`, which redirects the artifact directory — writes `masters.wl`, `recurrence_metadata.wl` (the per-sector M_1, M_0, diagonalizer and energy letters that feed the recurrence), `dlog_de.wl` and `manifest.wl` (version, master digest,

sector order, certification status and actual paths) to `results/madstree_formula/run-UUID/` by default. `MSEvaluatePath` and `MSReconstructEpSeries` automatically write or reuse these analytic files before numerical transport and return their paths under `"formulaArtifacts"`; the write is refused unless the d log-form DE is certified.

- Evaluation export: `MSExportEvaluationData[result, MSOutputDirectory -> ...]` writes `evaluation_data.csv` and `evaluation_data.json` under the caller-chosen output directory (default `results/madstree_evaluation/run-UUID/`), for downstream plotting tools.
- Cache identity: a successful-result cache key includes the request and the SHA-256 identities of the backend adapter, the configured backend's `pyproject.toml` and sorted `flintnde/*.py`, so in-place source fixes invalidate stale cache entries automatically.

The package source directory never receives run artifacts.

3.3 Numerical evaluation

With an initialized context, numerical evaluation proceeds in two steps: boundary generation and transport. The workflow is single-stage: `MSEvaluatePath` partitions the user point sequence into maximal consecutive complex-affine one-variable segments, pulls the d log-form connection back once per segment, and sends all segments, the exact boundary conditions and the user options to `FlintNDE` in one UTF-8 JSON request. With the default `FlintNDEPathPlanning -> True`, `FlintNDE` plans the transport nodes inside each segment and evaluates user points by fast multipoint evaluation; with `False` it uses every supplied point as a node in input order, without inserting, removing, or replanning points. The bundled backend is a synchronized copy of the standalone `FlintNDE` package released in the same repository; its mechanisms are described below. `MSBoundaryData` and `MSEvaluatePath` default to 200 decimal working digits; an explicit `WorkingPrecision` overrides the default.

Boundary generation.

```
targetRules = {k1 -> 9 I, k2 -> 3 I, q -> 1, a1 -> 1, a2 -> 1};

certificate = MSBoundaryChartCertificate[context, targetRules,
  RankOrder -> All];

boundary = MSBoundaryData[context, targetRules,
  BoundaryScale -> 4, WorkingPrecision -> 200];
```

`MSBoundaryChartCertificate` verifies the nested blow-up chart of Sec. 2.5 (normal crossing, theta fixing, quotient checks) for the chosen strict energy rank; `RankOrder -> All` checks every strict chart. `MSBoundaryData` consumes the certificate and returns the leading Frobenius data at the $K_v \to \infty$ boundary. For every master whose leading weight does

not vanish, it returns one branch record with the Frobenius exponent, the overall coefficient and the normalized leading vector, together with the anchor rules, the rank order and the certificate itself. The anchor rules normalize each leading vector: the entry of the anchoring master is fixed to one, entries of masters not reached by that solution are zero, and the remaining entries follow from a linear system that propagates leading subsector data into the masters of higher sectors. A master with a vanishing leading coefficient produces no record, and an unsolvable system is reported as a failure. If the chart is not certified, it returns `BoundaryChartNotCertified` and does not proceed to numerics. The energies in the example are positive imaginary so that the vertex factors $e^{-\mathrm{i}E\tau}$ damp at $\tau \to -\infty$; the rank order of the boundary is derived from the supplied target point, largest damping energy first, unless `RankOrder` fixes it.

Transport.

```
pointSequence = {{k1, k2}, {9 I, 3 I}};
parameterRules = {q -> 1, a1 -> 1, a2 -> 1};

result = MSEvaluatePath[context, pointSequence,
  ParameterRules -> parameterRules,
  BoundaryScale -> 4, RankOrder -> {v1, v2},
  WorkingPrecision -> 200, TransportOrder -> 80,
  ReferenceTransportOrder -> 104,
  TargetRelativeError -> "1e-20",
  MessageLanguage -> "EN"];

result["pointResults"]
```

The second argument is the *point sequence*: its first row lists the movable coordinate symbols, and every further row supplies one numerical value for each of them, in the same order. The remaining symbols of the target point, here `q`, `a1` and `a2`, are fixed once for the whole run through `ParameterRules`. For several target points, append one row per point,

```
pointSequence = {{k1, k2}, {8 I, 2 I}, {7 I, 4 I}, {6 I, 5 I}};
```

and `MadStree` groups the points into maximal consecutive complex-affine segments and pulls each segment back once, while `FlintNDE` evaluates every point in one process, including the extra points inside a node's convergence disk, without changing the path. For a single-vertex integral family, run the same single-stage entry on the vertex context. The principal transport options and the point tags are summarized in Sec. 3.5. If every d log-form letter is constant on a segment, every pulled-back derivative vanishes; the segment is then a zero connection with no finite poles and transport proceeds as for any ordinary segment.

Reading the result. `result` is an association whose `"pointResults"` entry is the list of point records. Every saved ordinary point carries `"coordinate"`, `"value"` (the master

vector in master order), `"status"` (equal to `"saved"` for ordinary points) and `"userIndex"`; transient `"tmp"` points are used only as intermediate points of the path. The current version does not provide leading-order point labels: every tag other than `"tmp"` is rejected. A user chain that crosses a pole or leaves a local convergence disk fails closed; the two options that decide this, `SingularityMode` and `FlintNDEPathPlanning`, are described in Sec. 3.5. The exported per-point `"relativeDifferenceInf"` and `"targetRelativeErrorMet"` are obtained from the truncation certification of Sec. 3.3. To read the first saved value, use

```
savedPoints = Select[result["pointResults"], #["status"] === "saved" &];
firstPoint = First[savedPoints];
firstPoint["value"]
```

The numerical backend. `FlintNDE` solves first-order matrix differential equations with exact $Q(\mathrm{i})(s)$-coefficient systems; `MadStree` embeds a synchronized copy and feeds it only single-variable pullbacks of the d log-form connection. Three mechanisms are responsible for the current performance:

- *Pole–residue transport.* An ordinary segment is serialized as its pole–residue representation $A(x) = P(x) + \sum_j R_j/(x - p_j)$, with P of arbitrary degree and the residues R_j of coincident letters merged. The backend generates the solution Taylor coefficients by a recurrence that carries one auxiliary state vector per pole, avoiding the Cauchy sampling and discrete-Fourier coefficient reconstruction of a generic analytic-matrix backend.

- *Embedded truncation certification.* The backend is always called in its embedded certification mode, which runs one chain at `ReferenceTransportOrder` and takes the primary result from the coefficient prefix truncated at `TransportOrder`, re-evaluated by Horner's rule. The per-segment truncation difference is returned as a local error estimate, so the `"relativeDifferenceInf"` and `"targetRelativeErrorMet"` fields remain available without the cost of a second chain. The call is upgraded to independent primary and reference chains whenever the prefix cannot carry the certification: on a multi-segment path, on a path with singular local data, for a run that starts from Frobenius boundary data, and for a request with saved points.

- *Fast multipoint evaluation.* With `FlintNDEPathPlanning -> True`, `FlintNDE` plans nodes inside each segment; user points covered by one expansion node form an evaluation bucket and are evaluated from the cached vector series. Buckets with at least eight points use the subproduct/remainder tree of fast multipoint evaluation, and small buckets use a simple iterative algorithm to avoid the overhead of the tree. User points in the same complex-affine segment $x(s) = x_0 + s\,v$ share node coefficients whenever they lie in the same convergence disk; different segments never share local coefficients.

Singularity handling is explicit: the default `SingularityMode -> "Automatic"` maps to the backend jump mode, so a segment that meets a supported singularity is crossed by an

explicit local-basis jump instead of being refused or detoured. The three modes and their interaction with `FlintNDEPathPlanning` are listed in Sec. 3.5. Exact Lee–Moser balances, strictly decoupled exponential times, and power-log recurrences are handled, while resonances whose local bases cannot be constructed, and any ramification or general Stokes requirement, fail closed. The internal working bits are $\lceil \mathrm{WP} \log_2 10 \rceil + 32$, where WP is the value of `WorkingPrecision` in decimal digits, and all path geometry (segment projections, match ratios, rotation factors, winding/monodromy) is kept in Arb/Acb precision. If the backend rewrites the problem in real variables, so that $k = -\mathrm{i}\kappa$, then $-k^2 = +\kappa^2$; a program token named `ik` then denotes the real variable κ, and `ik^2` is its square, not the imaginary unit times k^2.

3.4 Laurent-series reconstruction

When all vertex time powers share a common analytic regulator ε, e.g. $a_v = 1 + \varepsilon$, the master integrals develop a Laurent series

$$I(\varepsilon) = \sum_{p=n_{\min}}^{m} c_p \, \varepsilon^p + O(\varepsilon^{m+1}), \tag{3.2}$$

and `MSReconstructEpSeries[context, ep, pointSequence, MaximumEpPower -> m]` reconstructs the coefficients $c_{n_{\min}}, \ldots, c_m$. The user supplies no ε values; the lowest power is not a numerical fit parameter.

This procedure follows the one used by AMFlow [59, 60]. The transport is numerical, while the regulator dependence of the result has to stay analytic, and carrying that analytic dependence through every matrix multiplication of a long path would dominate the cost. Each sample point is therefore evaluated numerically at a fixed value of ε, and the coefficients of (3.2) are recovered afterwards from the final values of all runs, which is what keeps a multi-order expansion affordable.

Certification of the lowest power. Before any numerical solve at nonzero ε, `MadStree` constructs a support certificate from the symbolic boundary conditions and the d log-form DE of the production context. It requires:

1. regulator-independent path coordinates and no negative Laurent power at $\varepsilon = 0$ in any d log-form matrix, pulled-back connection, or residue;

2. finite-Laurent boundary branches, where branches with the same limiting Frobenius exponent and log degree are first combined with their physical weights and leading vectors, and only then are the component valuations of the record field `"componentValuations"` extracted;

3. at least one provably nonzero coefficient at the lowest power of the combined boundary; a noninteger power, non-Laurent structure, undecidable coefficient, or support entirely above the requested user range fails closed.

For $n_{\min} < 0$ the certificate describes a finite meromorphic Laurent boundary, not an analytic boundary at $\varepsilon = 0$; analyticity is recorded only when $n_{\min} \geq 0$. Along the current

finite path an ε-analytic invertible fundamental matrix preserves the valuation of nonzero formal boundary conditions, so the certified $n_{\min}$ is passed directly to the sampling plan. A rank drop of an individual Frobenius recurrence operator at $\varepsilon = 0$ is retained as a resonance diagnostic only; it is not by itself interpreted as a $1/\varepsilon$ pole of the already combined physical boundary.

Sampling, fitting and validation. Given the highest requested power m and the certified lowest power $n_{\min}$, the plan sets

$$P = \max(0, -n_{\min}), \qquad \Delta = m - n_{\min}, \qquad N_{\text{fit}} = \max\left(\left\lceil \tfrac{5}{2}\Delta + P \right\rceil, \Delta + 1 + f\right), \tag{3.3}$$

where $f =$ `EpFitExtraOrder`, default 2, is the number of powers the fit keeps above the user maximum: the second term in the maximum in (3.3) would be $\Delta + 1$ if only $c_{n_{\min}}, \ldots, c_m$ were fitted, and the extra f is what makes the interpolation reach past c_m in every run. A user-supplied pool `EpSamplePoints` and the refit rounds below may raise N_{fit} further, but N_{fit} is never below $\Delta + 1$. The plan then chooses the ε scale and precisions

$$\alpha_\varepsilon = \frac{P}{4} + \frac{g}{\Delta + 1}, \qquad \varepsilon_0 = 10^{-\alpha_\varepsilon}, \qquad p_0 = \max(\lceil (N_{\text{fit}} + P)\alpha_\varepsilon \rceil, 30), \tag{3.4}$$

where $g =$ `EpGoalDigits` sets the validation tolerance 10^{-g}. The working precision is $\max(200, \lceil 2(1+e)p_0 \rceil)$ with e the extra-precision fraction; the primary transport order is $4p_0$ and the reference chain adds $\lceil \max(50, p_0/5) \rceil$ orders. All N_{fit} production points enter one square Acb Vandermonde system with powers up to $\varepsilon^{n_{\min}+N_{\text{fit}}-1}$; only $c_{n_{\min}}, \ldots, c_m$ are returned, and no least squares or pseudoinverse is used. Independent validation points lie at a fixed fraction of the production scale and never enter the fit. At each point, the fitted series includes all N_{fit} internal powers through $n_{\min} + N_{\text{fit}} - 1$, including the buffer powers above the requested maximum m. This sum must reproduce the directly transported value to the relative tolerance 10^{-g}. If the target is not met after the allowed refit rounds, the result carries a warning status and `"precisionTargetMet" -> False`. The residual tests the fitted series at the validation points; it does not establish the number of correct digits of each returned coefficient $c_{n_{\min}}, \ldots, c_m$.

Incremental expansion and parallelism. The initial fit includes at least two powers above the user maximum (`EpFitExtraOrder -> 2`). If validation fails, each subsequent round adds `EpFitOrderIncrement -> 2` powers, solves only the new production points, and reuses every previous production and validation value, for at most `EpFitMaximumRounds -> 3` rounds; the tolerance is not relaxed and the caches are not discarded. Independent ε tasks run in an outer process pool bounded by `ParallelTaskCount -> 12`, whose effective concurrency is the smaller of the limit and the number of tasks; this is separate from the in-process `ctx.threads` of `python-flint`. The third application example in Sec. 4.2 runs this procedure on a divergent configuration.

3.5 Options, validation and common pitfalls

The principal options of the numerical workflow are summarized in Table 3.5.

option	meaning
`FlintNDEPathPlanning`	`True` (default): `FlintNDE` plans nodes and uses fast multipoint evaluation; `False`: user points as nodes in input order
`SingularityMode`	`"Automatic"` (default), `"Avoid"` or `"SingularityJump"`
`MessageLanguage`	`"EN"` (default) or `"CN"`, case-sensitive
`BoundaryScale`	scale setting the damping energies at the finite boundary anchor (> 1; default 8)
`BoundarySeriesOrder`	boundary-order metadata and regulator recurrence diagnostics; not the numerical Frobenius start order in the present generic workflow
`RankOrder`	vertex order by damping energy K_v (default `Automatic`: from the first target point)
`WorkingPrecision`	integer working precision in decimal digits, at least 20; default 200
`TransportOrder` / `ReferenceTransportOrder`	primary and reference local-series orders, including the numerical Frobenius start; defaults 48 and 64, with the reference order strictly larger
`TargetRelativeError`	target relative error of the primary/reference check; default `"1e-25"`
`MaximumEpPower`	highest returned regulator power of `MSReconstructEpSeries` (default 0)
`EpGoalDigits`	decimal tolerance exponent for validation of the reconstructed regulator series (default 20)
`EpFitExtraOrder` / `EpFitOrderIncrement` / `EpFitMaximumRounds` / `EpInitialInternalMaximumPower`	internal buffer powers, per-round increment, maximum refit rounds and first-round internal highest power of the regulator fit (defaults 2, 2, 3, `Automatic`)
`ParallelTaskCount`	maximum concurrent `ep` backend processes in `MSReconstructEpSeries` (default 12)
`PythonExecutable`	Python command for the `FlintNDE` adapter (default `Automatic`)
`MSRuntimeDirectory`	temporary-file root (default: `results_temp` in the calling script or notebook directory)

Table 3.5. Principal numerical options and their defaults.

The transport orders and `TargetRelativeError` in Table 3.5 apply to `MSEvaluatePath`. For `MSReconstructEpSeries`, these three options are not accepted; the numerical settings are planned from `EpGoalDigits`. Its `BoundarySeriesOrder` defaults to `Automatic`, whereas `MSEvaluatePath` uses 24.

The saved/transient distinction is not a transport option: it is a tag attached to the coordinate rows below the symbol header, which are bare value lists or {{`values...`},`"tmp"`}. With `SingularityMode -> "Avoid"`, interior singularity crossings are refused, and the

refusal names the interior singular points of the path. The default `"Automatic"` mode allows supported singularity jumps when `FlintNDEPathPlanning -> True`; an explicit `"SingularityJump"` also requires this setting. When planning is disabled, `"Automatic"` behaves as `"Avoid"`. An explicitly requested terminal singular point is handled by local classification and evaluation, rather than by ordinary Taylor transport through the singularity.

Limitations and fail-closed guarantees:

- No loop-momentum integration, no sampled IBP, no Kira or external reduction.
- `"dlogStatus"` must be `"certifiedByFormulaChecks"`; otherwise the DE is not consumed.
- Uncertified boundary charts and singular points beyond the backend's certified capabilities all fail closed; resonances whose local bases cannot be constructed return structured refusals, and ordinary Taylor transport is never used across singular points.
- An incomplete `MSReduce` call may return an association with `"status" -> "partiallyReduced"` or `"status" -> "failed"`, rather than a top-level `Failure`. Inspect `"remainingShiftedIntegrals"` and `"nonMasterResidual"`.
- For `MSEvaluatePath`, check `result["flintNDE"]["targetRelativeErrorMet"]`, not only `result["status"]`. For `MSReconstructEpSeries`, check `"precisionTargetMet"`; coefficients may be returned with `"status" -> "computed_with_warning"` if the target is not met.
- The production boundary never uses defining integrals; finite-point defining integrals exist only inside independent validation programs.

Common pitfalls.

1. **Convention**: `NuConvention` is chosen at initialization only (`"Positive"` by default). To compare with the formulas of [43], initialize with `NuConvention -> "Negative"`, which selects the minus sign in (3.1).
2. **Sector keys**: fixed-length strings; leading zeros and the string type are part of the identity (Sec. 2.3).
3. **Single-stage workflow**: call `MSEvaluatePath` with a coordinate table and fixed `ParameterRules`. The call generates the boundary data once and transports the solution along the full point sequence.
4. **Singularity handling**: the default is `SingularityMode -> "Automatic"`, where `"Automatic"` is a string, not the symbol `Automatic`. Use `"Avoid"` to refuse interior crossings. Supported jumps require `FlintNDEPathPlanning -> True`, and the user must check the multivalued branch.

5. **Point tags**: after the coordinate-symbol header, only bare value lists and {{`values...`},`"tmp"`} are accepted; every other tag, including `"lo"`, is rejected.

6. **Backend environment**: `MadStree` follows the zero-probe policy of Sec. 3.1, so the interpreter is the one selected there and must be able to import the embedded `flintnde`. Backend failures report captured output through messages such as `backendLaunchFailed` and `backendRunFailed`; retained runtime logs contain the details. After a successful call, and after a structured refusal of a singular path, the package removes the temporary input and log files.

7. **Message language**: `MessageLanguage` accepts only `"EN"` and `"CN"` (case-sensitive), with `"EN"` the default.

8. **Encoding**: loading through `Needs` uses explicit UTF-8 internally and needs no user encoding option; if loading the kernel file directly fails with syntax errors, set `$CharacterEncoding = "UTF-8"` before loading.

4 Examples

This section lists the examples distributed with the package and three applications computed with them. Sec. 4.1 describes the seven distributed examples, which exercise recurrence reduction, single-vertex families, time-order charts, massless and massive trees, zero external-leg energy, and the regulator reconstruction of Sec. 3.4. Sec. 4.2 presents three numerical applications: the tree with three vertices and two massive exchanges, whose transported results are compared with the multivariate hypergeometric representation of [3]; the bubble time kernel evaluated at fixed internal momenta; and the massless three-vertex chain evaluated in the regulator mode, which demonstrates the certified Laurent reconstruction of Sec. 3.4 on a divergent configuration. The comparisons underlying the verification tasks of Sec. 5 are not repeated here. They cover the single-vertex families and the massive two-vertex correlator against [43], and the IBP relations and differential equations against the `dSIBP` package [63].

4.1 Package examples

Each example is a Wolfram Language script; together they exercise the complete workflow:

- `01_massless_full_edge.wl`: the massless shared mode-function two-state simplification of Sec. 2.2, masters, recurrence, d log-form DE and the automatic boundary/numerical entry; it also demonstrates the single-stage multipoint workflow, with CSV/JSON export;

- `02_single_vertex_family.wl`: single-vertex integral family input, local tensor inverses (Sec. 2.2), and finite-combination reduction;

- `03_time_only_cycle_chart.wl`: time-order cycle input, common-theta contacts (Sec. 2.6 and App. A), and all strict-rank chart certificates; contacts are discussed in Sec. 2.3;

- `04_three_vertex_tree.wl`: a three-vertex, two-propagator massless tree (+++ vertex structure), following the same flow as example 01, with a single-stage multipoint evaluation and CSV/JSON export;

- `05_massive_three_vertex_tree.wl`: a three-vertex, two-propagator massive tree (+ + + vertex structure) with non-half-integer `nu12=3/4`, `nu23=1/3`, followed by a single-stage multipoint evaluation and CSV/JSON export;

- `06_massless_three_vertex_ep_regularization.wl`: a three-vertex massless tree with the analytic regulator $a_1 = -1 + \varepsilon$, $a_2 = a_3 = 1 + \varepsilon$, whose $\varepsilon = 0$ time integrals genuinely diverge; before the numerical stage it reduces the sum of two integrals whose time powers lie one unit below the master powers (displayed in Sec. 4.2); the user requests only the highest regulator power (`MaximumEpPower -> 0`), and the package certifies the lowest power $n_{\min}$ from the symbolic boundary and DE before any numerical solve (text below); its numerical reconstruction is presented as the third application in Sec. 4.2;

- `07_zero_external_leg_energy.wl`: zero external-leg energy, represented through an auxiliary energy with zero target value.

In Example 06, the vertex-1 unit $u_1 = a_1 + 1$ equals ε, and every factorized boundary weight that isolates this unit as a standalone factor carries a $\Gamma(\varepsilon)$ simple pole, while the nine-dimensional DE has no negative ε power. The certificate therefore gives $n_{\min} = -1$, decided by the combined physical boundary and not by any individual resonance.

Each example ends with a failure gate: it prints `Example PASSED` and exits 0 only when every checked result is a non-`Failure` object, and exits 1 with the reason otherwise. Run them in a Mathematica front end section by section, or with

```
wolframscript -file .../Examples/01_massless_full_edge.wl
```

4.2 Application examples

We illustrate the workflow of `MadStree` with three examples, using `FlintNDE` for numerical transport. All timings measure elapsed time on a personal computer.

Tree with three vertices and two massive scalar exchanges The first example is the chain with three vertices in figure 2.

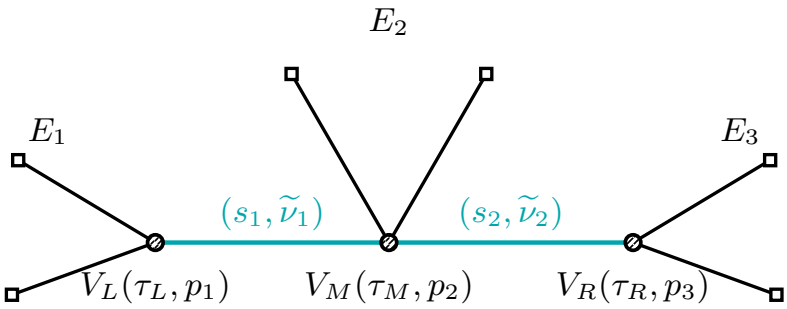


Figure 2. Tree with three vertices and two massive scalar propagators, drawn with the in-in Feynman rules of Sec. 2.1 for the all-plus assignment $(+++)$: the time vertices are hatched circles on a horizontal axis, all on the `"+"` branch of the in-in contour, each carrying its conformal time τ_L, τ_M or τ_R and time power p_1, p_2, p_3 in the label $V_v(\tau_v, p_v)$ below it, together with a schematic fan of two external legs ending in open squares and labeled by the summed external energy $E_v = \sum_j k_{v,j}$. The internal propagators are drawn in teal and carry their momentum magnitudes and mass labels $(s_1, \widetilde{\nu}_1)$, $(s_2, \widetilde{\nu}_2)$ above the propagator.

The vertices L, M, R have conformal times τ_L, τ_M, τ_R. The two internal propagators connect L to M and M to R, respectively. Each vertex carries a fan of external legs; the momentum magnitudes $k_{v,j}$ of the legs at vertex v sum to its external energy, $E_v = \sum_j k_{v,j}$. The internal momentum magnitudes are s_1, s_2, and the Hankel orders are ν_1, ν_2. For the masses in the principal series used here, $\nu_i = \mathrm{i}\widetilde{\nu}_i$, with real $\widetilde{\nu}_i$. Each vertex has a contour sign $\sigma_v = \pm 1$; figure 2 is drawn for the all-plus assignment $\sigma_L = \sigma_M = \sigma_R = +1$, and the assignments computed in this application are fixed below.

This application uses `NuConvention -> "Negative"` to match [43]. The simultaneous replacement $\nu \to -\nu$ relative to (2.8) applies throughout, including the effective vertex time powers below. From figure 2, the function family of this example is built from one time integration per vertex, carrying the time power $(-\tau_v)^{\nu_{0v}}$ and the external exponential $e^{\mathrm{i}\sigma_v E_v \tau_v}$, and from one propagator kernel per internal propagator, each kernel carrying one mode-function state at each of its two vertices. We first make these mode-function states explicit:

$$\begin{aligned} h^{(r)}(\nu, 0; z) &= z^{-\nu} H^{(r)}_{\nu}(z), \\ h^{(r)}(\nu, 1; z) &= \partial_z h^{(r)}(\nu, 0; z) = -z^{-\nu} H^{(r)}_{\nu+1}(z), \qquad r = 1, 2. \end{aligned} \tag{4.1}$$

Here r selects the Hankel function and the binary index selects the undifferentiated or differentiated mode-function state. We use the same Hankel order at both vertices of each propagator; constant factors from the physical mode normalization are restored when forming the correlator. To display the time ordering explicitly, define

$$\begin{aligned} W_{>,\nu;ab}(s; \tau, \tau') &= h^{(1)}(\nu, a; -s\tau) h^{(2)}(\nu, b; -s\tau'), \\ W_{<,\nu;ab}(s; \tau, \tau') &= h^{(2)}(\nu, a; -s\tau) h^{(1)}(\nu, b; -s\tau'), \qquad a, b \in \{0, 1\}. \end{aligned} \tag{4.2}$$

These two products are $G_>$ and $G_<$ of (2.2) with the mode prefactors removed and the two mode-function states made explicit. The propagator kernel of contour type $\sigma\sigma'$, written

$\mathcal{G}_{\sigma\sigma',\nu;ab}(s;\tau,\tau')$ below, is then the time-ordered combination of the two products exactly as (2.3) combines $G_>$ and $G_<$: $\mathcal{G}_{++}$ and $\mathcal{G}_{--}$ take $\theta(\tau-\tau')W_>$ plus $\theta(\tau'-\tau)W_<$ and $\theta(\tau-\tau')W_<$ plus $\theta(\tau'-\tau)W_>$, while $\mathcal{G}_{+-} = W_{<,\nu;ab}$ and $\mathcal{G}_{-+} = W_{>,\nu;ab}$ carry no theta function. The binary indices act only on the Hankel factors, never on the step functions, and every kernel has the arguments $(s;\tau,\tau')$.

For $\boldsymbol{\sigma} = (\sigma_L, \sigma_M, \sigma_R)$, the Top-sector master integrals are

$$\begin{aligned} \mathrm{I}^{\boldsymbol{\sigma}}_{abcd} = \int_{-\infty}^{0} d\tau_L\, d\tau_M\, d\tau_R\, (-\tau_L)^{\nu_{0L}} (-\tau_M)^{\nu_{0M}} (-\tau_R)^{\nu_{0R}} \\ \times e^{\mathrm{i}(\sigma_L E_1 \tau_L + \sigma_M E_2 \tau_M + \sigma_R E_3 \tau_R)} \\ \times \mathcal{G}_{\sigma_L\sigma_M,\nu_1;ab}(s_1;\tau_L,\tau_M)\mathcal{G}_{\sigma_M\sigma_R,\nu_2;cd}(s_2;\tau_M,\tau_R). \end{aligned} \tag{4.3}$$

Here $a,b,c,d \in \{0,1\}$. Each time integral runs from $-\infty$ to 0 with the Bunch–Davies prescription. Each massive propagator carries one mode-function state at each of its two vertices. The four mode-function state indices a,b,c,d are ordered (L_1, M_1, M_2, R_2): a,b are the states of propagator s_1 at τ_L, τ_M and c,d those of s_2 at τ_M, τ_R. Thus the Top sector consists of the sixteen masters

$$\begin{aligned} \mathrm{I}^{\boldsymbol{\sigma}}_{\mathrm{Top}} = \big(&\mathrm{I}^{\boldsymbol{\sigma}}_{0000}, \mathrm{I}^{\boldsymbol{\sigma}}_{0001}, \mathrm{I}^{\boldsymbol{\sigma}}_{0010}, \mathrm{I}^{\boldsymbol{\sigma}}_{0011}, \mathrm{I}^{\boldsymbol{\sigma}}_{0100}, \mathrm{I}^{\boldsymbol{\sigma}}_{0101}, \mathrm{I}^{\boldsymbol{\sigma}}_{0110}, \mathrm{I}^{\boldsymbol{\sigma}}_{0111}, \\ &\mathrm{I}^{\boldsymbol{\sigma}}_{1000}, \mathrm{I}^{\boldsymbol{\sigma}}_{1001}, \mathrm{I}^{\boldsymbol{\sigma}}_{1010}, \mathrm{I}^{\boldsymbol{\sigma}}_{1011}, \mathrm{I}^{\boldsymbol{\sigma}}_{1100}, \mathrm{I}^{\boldsymbol{\sigma}}_{1101}, \mathrm{I}^{\boldsymbol{\sigma}}_{1110}, \mathrm{I}^{\boldsymbol{\sigma}}_{1111}\big)^{\mathsf{T}}, \end{aligned} \tag{4.4}$$

one for each of the sixteen values of $(a,b,c,d) \in \{0,1\}^4$. The entry $\mathrm{I}^{\boldsymbol{\sigma}}_{0000}$ is the one with no derivative acting on the mode-function states and no additional integer shifts in the time powers. Equation (4.3) is the sector Top $= (\xi_1\xi_2) = (11)$ of (2.14), with $c_{\mathrm{Top}} = c_V = 3$ components by (2.15) and $n^{\mathrm{slot}} = 4$ slots, and

$$\mathrm{I}^{\boldsymbol{\sigma}}_{abcd} = \mathrm{I}_{\mathrm{Top}}\big(\mathbf{0}; (a,b,c,d)\big), \tag{4.5}$$

the bit order of (2.12) being the one used by $\mathrm{I}^{\boldsymbol{\sigma}}_{\mathrm{Top}}$.

The parameters $\nu_{0L}, \nu_{0M}, \nu_{0R}$ in the integrand are the effective vertex time powers. If p_1, p_2, p_3 are the powers before absorbing the massive mode prefactors, then

$$\begin{aligned} \nu_{0L} &= p_1 + \frac{3}{2} + \nu_1, \qquad \nu_{0R} = p_3 + \frac{3}{2} + \nu_2, \\ \nu_{0M} &= p_2 + 3 + \nu_1 + \nu_2. \end{aligned} \tag{4.6}$$

The vertex factors $\mathrm{i}\sigma_v$ and the external and mode normalizations are not included in (4.3); they are restored when constructing the physical correlator.

In the IBP reduction, derivatives of the functions that impose time ordering generate contact terms that merge adjacent vertices. These terms define the LeftPinch, RightPinch and DoublePinch sectors. The left pinch merges L and M and leaves the two indices of propagator 2, so its four masters are the child integrals of the contact term (2.49), written

out as

$$\begin{aligned}\mathrm{I}^{\boldsymbol{\sigma},\mathrm{L}}_{cd} = &\int_{-\infty}^{0}\mathrm{d}\tau_{LM}\int_{-\infty}^{0}\mathrm{d}\tau_{R}\,(-\tau_{LM})^{\nu_{0L}+\nu_{0M}-2\nu_1}(-\tau_{R})^{\nu_{0R}}\\ &\times e^{\mathrm{i}(\sigma_L E_1+\sigma_M E_2)\tau_{LM}+\mathrm{i}\sigma_R E_3\tau_R}\,\mathcal{G}_{\sigma_M\sigma_R,\nu_2;cd}(s_2;\tau_{LM},\tau_R), \qquad c,d\in\{0,1\},\end{aligned} \tag{4.7}$$

where τ_{LM} is the single time of the merged component and the merged power is $A_{C,\mathrm{L}}$ of (2.41): the raw contact power of the contracted propagator is $\varrho_1 = -2\nu_1 - 1$, and the component count $|C| = 2$ adds the remaining $+1$, so that (4.6) gives $\nu_{0L} + \nu_{0M} - 2\nu_1 = p_1 + p_2 + \frac{9}{2} + \nu_2$. The right pinch merges M and R and keeps the two indices of propagator 1; its four masters are obtained from (4.7) by the exchange $(L, \nu_{0L}, \sigma_L E_1, s_1, \nu_1) \leftrightarrow (R, \nu_{0R}, \sigma_R E_3, s_2, \nu_2)$,

$$\begin{aligned}\mathrm{I}^{\boldsymbol{\sigma},\mathrm{R}}_{ab} = &\int_{-\infty}^{0}\mathrm{d}\tau_{L}\int_{-\infty}^{0}\mathrm{d}\tau_{MR}\,(-\tau_{L})^{\nu_{0L}}(-\tau_{MR})^{\nu_{0M}+\nu_{0R}-2\nu_2}\\ &\times e^{\mathrm{i}\sigma_L E_1\tau_L+\mathrm{i}(\sigma_M E_2+\sigma_R E_3)\tau_{MR}}\,\mathcal{G}_{\sigma_L\sigma_M,\nu_1;ab}(s_1;\tau_L,\tau_{MR}), \qquad a,b\in\{0,1\},\end{aligned} \tag{4.8}$$

where τ_{MR} is the time of the merged M, R component and $\nu_{0M}+\nu_{0R}-2\nu_2 = p_2+p_3+\frac{9}{2}+\nu_1$. The double pinch merges all three vertices and leaves no massive propagator, so its single master is the one-time integral

$$\mathrm{I}^{\boldsymbol{\sigma},\mathrm{D}} = \int_{-\infty}^{0}\mathrm{d}\tau\,(-\tau)^{\nu_{0L}+\nu_{0M}+\nu_{0R}-2\nu_1-2\nu_2}e^{\mathrm{i}(\sigma_L E_1+\sigma_M E_2+\sigma_R E_3)\tau}, \tag{4.9}$$

whose power is $p_1 + p_2 + p_3 + 6$. In the index convention (2.13) the four sectors are Top $=$ $(\xi_1\xi_2) = 11$ with three components and four slots, $(\mathrm{LeftPinch}, \mathrm{RightPinch}) = (01, 10)$ with two components and two slots each, and DoublePinch $= 00$ with one component and no slot, exactly as (2.15) requires of a graph with $c_V = 3$ vertices. When both internal propagators join vertices with equal contour signs, the full hierarchy is

$$\mathrm{Top}(16)\oplus\mathrm{LeftPinch}(4)\oplus\mathrm{RightPinch}(4)\oplus\mathrm{DoublePinch}(1), \qquad \dim = 25, \tag{4.10}$$

and the four sectors are shown in figure 3.

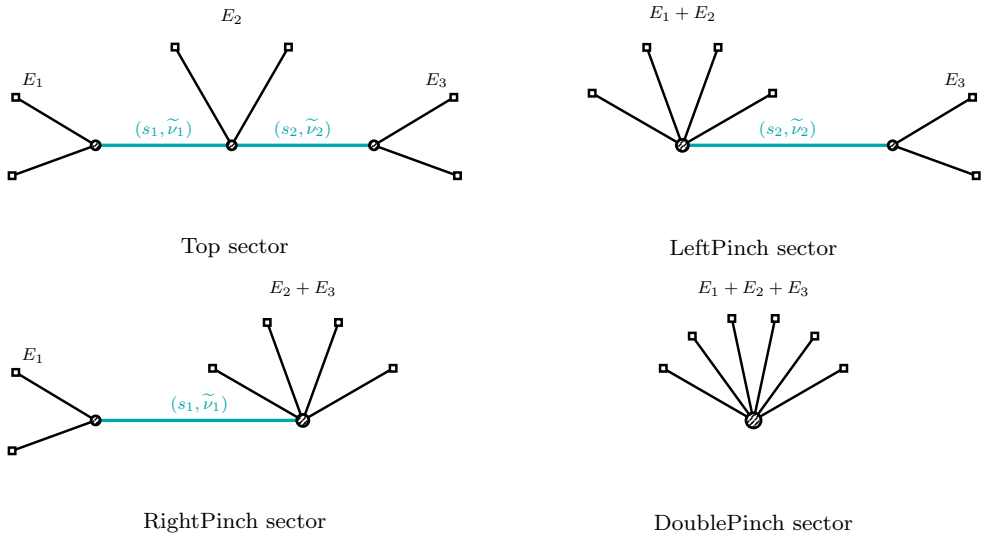


Figure 3. The four sectors of the three-vertex chain of figure 2, drawn in the same visual language. Contracting propagator 1 merges L and M (LeftPinch), contracting propagator 2 merges M and R (RightPinch), and both contractions merge all three vertices (DoublePinch). A merged component is drawn as a larger hatched circle carrying the external-leg fans of all vertices it contains, and the contracted propagator is absent; the surviving propagator keeps its mode-function two-state (massive chain) or shared (massless chain) slot and its teal label. All panels are drawn for equal contour signs. The massless chain of the regulator example (figure 9) carries the same four sectors.

For other contour assignments, a propagator whose two vertices carry opposite signs is theta-free and cannot be contracted, so only the allowed pinched sectors appear.

We write σ_{LM} and σ_{MR} for the contour signs of the two internal propagators, $+1$ for G_{++} and -1 for G_{--} as in (2.22). The lowering contact term of the Top IBP at $v = M$ is then the two-vertex relation (2.49), written in the indices of this graph; the contact at $v = L$ has the opposite sign,

$$\mathcal{R}^{(0)}_{v;\tilde{\boldsymbol{r}},\sigma_{LM}} = -\sigma_{LM}\,\delta_{a,1-b}\,(-1)^{a+1}\,\frac{4\mathrm{i}}{\pi}\,s_1^{-2\nu_1-1}\,\mathrm{I}_{\mathrm{L}}\big(-\boldsymbol{e}_{LM};(c,d)\big), \qquad \tilde{\boldsymbol{r}} = 1+2c+d, \quad (4.11)$$

where the child is taken at shift -1 on the merged component and 0 on τ_R, so that its merged power is one below (4.7), and the bits a, b of the contracted propagator are removed from the parent string, leaving $\tilde{\boldsymbol{r}}$ as the packed label of (c, d). The four sectors then carry the hidden constants

$$\begin{aligned} \mathcal{N}_{\mathrm{Top}} &= 1, & \mathcal{N}_{\mathrm{L}} &= -\sigma_{LM}\,\frac{4\mathrm{i}}{\pi}\,s_1^{-2\nu_1-1}, \\ \mathcal{N}_{\mathrm{R}} &= -\sigma_{MR}\,\frac{4\mathrm{i}}{\pi}\,s_2^{-2\nu_2-1}, & \mathcal{N}_{\mathrm{D}} &= -\sigma_{LM}\sigma_{MR}\,\frac{16}{\pi^2}\,s_1^{-2\nu_1-1}s_2^{-2\nu_2-1}, \end{aligned} \quad (4.12)$$

which is the product rule (2.16) with $\mathcal{N}^{(0)}_s = 1$. The object the package propagates in sector s is $J_s(\boldsymbol{n};\boldsymbol{a}) = \mathcal{N}_s\,\mathrm{I}_s(\boldsymbol{n};\boldsymbol{a})$ — concretely $J^{\boldsymbol{\sigma}}_{abcd} = \mathrm{I}^{\boldsymbol{\sigma}}_{abcd}$, $J^{\boldsymbol{\sigma},\mathrm{L}}_{cd} = \mathcal{N}_{\mathrm{L}}\mathrm{I}^{\boldsymbol{\sigma},\mathrm{L}}_{cd}$, $J^{\boldsymbol{\sigma},\mathrm{R}}_{ab} = \mathcal{N}_{\mathrm{R}}\mathrm{I}^{\boldsymbol{\sigma},\mathrm{R}}_{ab}$ and $J^{\boldsymbol{\sigma},\mathrm{D}} = \mathcal{N}_{\mathrm{D}}\mathrm{I}^{\boldsymbol{\sigma},\mathrm{D}}$ — and for the $(+++)$ assignment (4.12) is the value of the

`"normalization"` field of each master record.

At real kinematic points, we compute the four assignments $(+++)$, $(++-)$, $(+-+)$ and $(-++)$. After restoring the vertex factors and mode normalizations, each $\mathrm{I}^{\sigma}_{0000}$ entry gives one contribution to the physical correlator. Reversing all three contour signs gives the complex conjugate contribution, so the four computed terms and their conjugates determine $\mathcal{A}_{\mathrm{phys}}$. To match the phase convention above, the application code passes $-E_v$ as `externalLegEnergy`, uses `NuConvention -> "Negative"`, and supplies the corresponding pinch normalizations.

We set $p_1 = p_2 = p_3 = 0$ and express all energies and momenta in units of E_2. With $q = s_1/E_2$, the scan follows

$$\begin{aligned}(E_1, E_2, E_3, s_1, s_2) &= \left(\frac{100q}{99}, 1, \frac{10}{99}, q, \frac{1}{10}\right), \qquad (\nu_1, \nu_2) = (\mathrm{i}, 2\mathrm{i}),\\ (\nu_{0L}, \nu_{0M}, \nu_{0R}) &= \left(\frac{3}{2} + \mathrm{i}, 3 + 3\mathrm{i}, \frac{3}{2} + 2\mathrm{i}\right).\end{aligned} \tag{4.13}$$

As the numerical benchmark, which is independent of the DE construction, we use the multivariate hypergeometric representation of this correlator derived in [3]. It expresses the signal and background contributions as multiple series containing Appell F_2 functions. Their sum gives the complete physical correlator. These contributions are not the IBP sectors in (4.10). For this benchmark, DE transport evolves the full master vector for each contour assignment and then extracts the physical correlator from its $\mathrm{I}^{\sigma}_{0000}$ entries. The hypergeometric calculation evaluates the same physical correlator directly at each target point; it does not separately evaluate all entries of the master vectors. The reported comparison is therefore between physical correlators, not an entry-by-entry comparison of all masters. The truncation order is selected by comparing adjacent orders. The reported hypergeometric time is the sum of the accepted pointwise evaluation times.

The DE calculation uses 80 decimal digits and local Taylor series truncated at order 80. An independent transport at order 96 provides an internal stability check. We compare batches of $N = 1$, 10, 50 and 100 target points on the slice in (4.13). For $N = 1$, 10 and 50, transport starts at $q = 0.1$ and ends at $q = 10^{-4}$. The single point is the end of the path, corresponding to $x = -\log_{10} q = 4$. The 10 points are selected at approximately equal index intervals from the 50 points of the $N = 50$ batch, including both ends. The 100-point batch contains 80 points on this path and 20 additional points from $q = 0.105$ to 0.2. The additional points share the boundary at $q = 0.1$ but require transport in the opposite direction.

The hypergeometric times for these four batches are 22.78, 344.62, 1040.95 and 4242.89 s. The corresponding median DE times are 26.71, 48.82, 48.86 and 57.56 s (figure 4). Each DE time covers data preparation, the Python invocation, boundary construction, DE generation and the transport call, with the order-96 backend time subtracted. All four DE batches were measured three times, with the execution order varied between rounds. We report the median total time and show the full range of the three measurements. The observed ratios of hypergeometric time to DE time are 7.06, 21.31 and 73.71

for $N = 10$, 50 and 100, respectively. For the single-point batch at the far end of the path, DE transport is slower than the hypergeometric evaluation.

The average DE time per output point decreases from 26.71 s to 4.88, 0.98 and 0.58 s across these batches. Boundary data and transport are shared, rather than recomputed independently for each target. Points encountered along the path use saved node values or evaluations of an already constructed local series. In these runs, each branch used 29 internal nodes for one output point and 33 nodes for 10 output points. Thus, additional outputs did not require a proportional increase in transport work.

Within a convergence disk, a shorter step suppresses higher Taylor terms. It can therefore permit a lower truncation order at a fixed accuracy. This is a local property, not a guarantee of lower total cost, since shorter steps can require more segments.

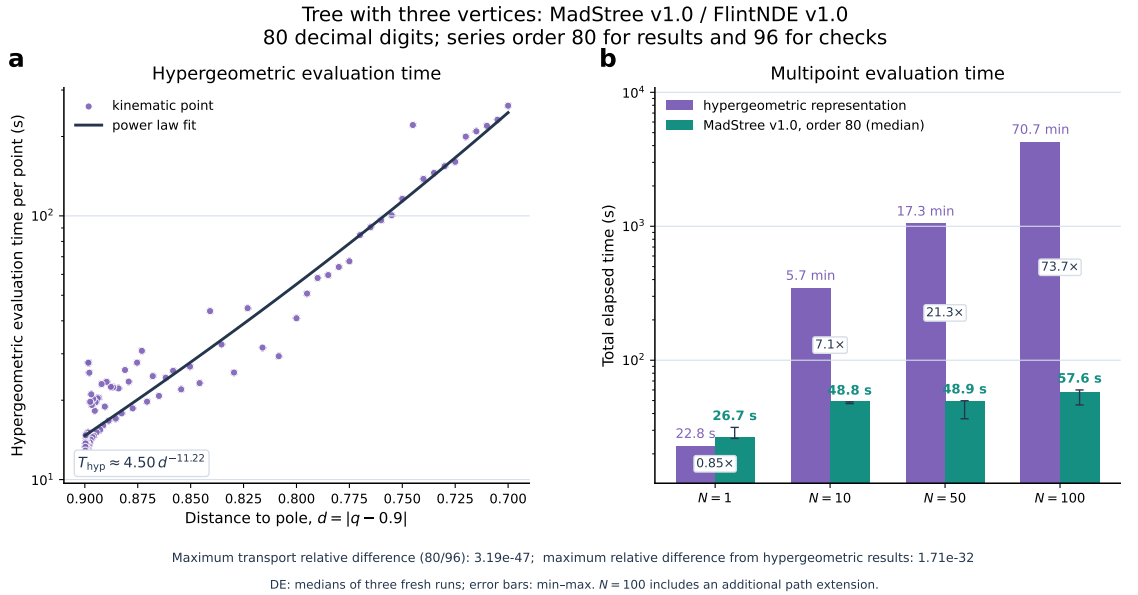


Figure 4. Tree benchmark with `MadStree` and `FlintNDE`. **a**, Hypergeometric evaluation time at individual kinematic points. **b**, Total evaluation times for $N = 1$, 10, 50 and 100 target points.

The 100 points in figure 4**a** comprise 80 points with $q = 10^{-x}$ and $x = 1 + 3j/79$ ($j = 0, \ldots, 79$), uniformly spaced in x, and 20 points with $q = 0.1 + 0.005j$ ($j = 1, \ldots, 20$), uniformly spaced in q. Each hypergeometric time is measured at the accepted truncation order. In panel **b**, the DE bars show medians of three runs, with error bars spanning the minimum and maximum. The first three batches share the same start and end; the 100-point batch includes the additional path extension. Hypergeometric data are newly computed for $N = 1$ and 10 and retained for $N = 50$ and 100. The ratio labels give the hypergeometric time divided by the DE time. The figure footer reports the maximum relative differences between the two transport orders, and between the DE and hypergeometric results, across all four batches.

Across the scan of 100 points, the maximum relative difference from the hypergeometric result is 1.712×10^{-32}. The comparison of transport orders 80 and 96 gives a maximum

relative difference of 1.308×10^{-50}. At $x = 4$, the physical correlator is

$$\mathcal{A}_{\text{phys}}(4) = 9.87033514798954 \times 10^{-9}, \tag{4.14}$$

with a relative difference of 4.21×10^{-36} from the hypergeometric result.

For the same kinematic configuration, the direct Appell F_2 representation requires $q < 0.89$. The point $q = 0.8905$ lies outside this domain, whereas the nearest positive pole of the DE connection is at $q = 0.9$. To reach $q = 1$, we continue the chosen analytic branch along

$$0.1 \to 0.86 \to 0.86 + 0.04\mathrm{i} \to 0.93 + 0.04\mathrm{i} \to 0.93 \to 1. \tag{4.15}$$

This contour avoids the pole through the upper half of the complex plane. The continuation records 81 points. The calculation at order 80 takes 35.08 s, or 45.76 s when the reference calculation at order 96 is included. The maximum relative difference between orders 80 and 96 is 1.802×10^{-44}. The DE method thus evaluates the selected analytic branch beyond the domain of the direct series representation.

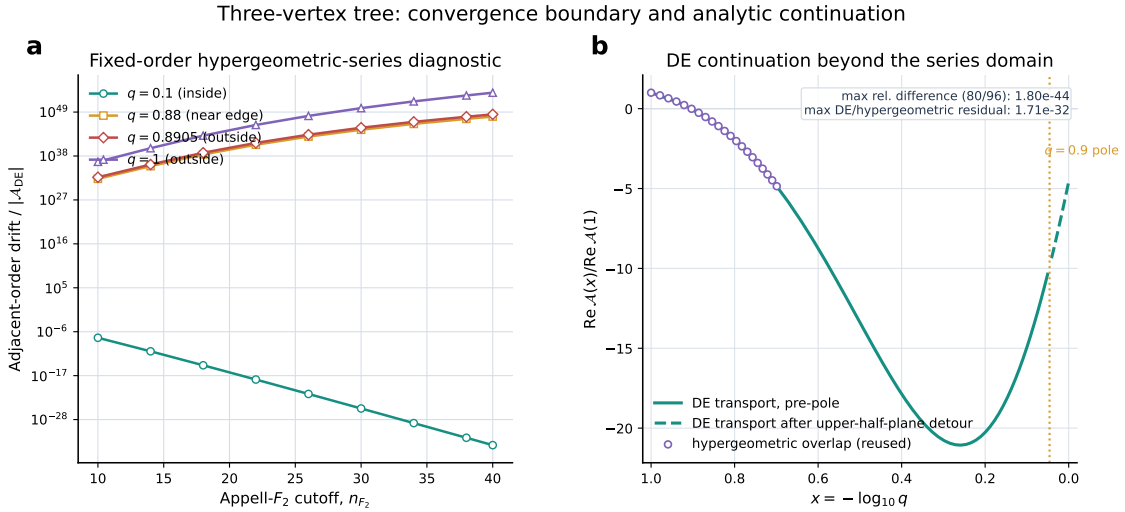


Figure 5. Analytic continuation with `MadStree`. **a**, Dependence of the hypergeometric partial sums on the truncation order near the direct convergence boundary. **b**, DE continuation at order 80 along the contour in (4.15).

Bubble time kernel at fixed loop momentum The second example has two time vertices joined by two massive propagators (figure 6). We evaluate the integrals over the two vertex times while holding the internal momenta fixed. We do not integrate over the spatial loop momentum.

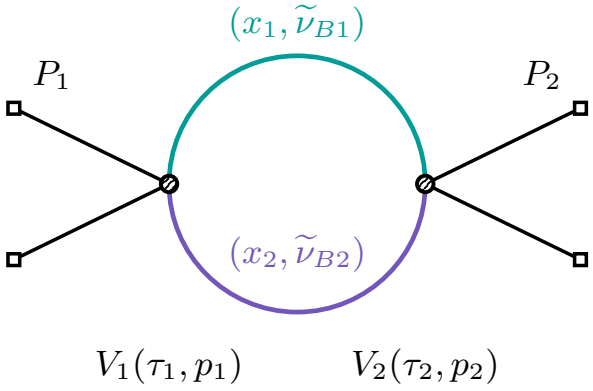


Figure 6. Bubble time kernel, drawn with the in-in Feynman rules of Sec. 2.1 for the assignment *pp*: both vertices are hatched circles on the "+" branch and carry the times τ_1, τ_2 and powers p_1, p_2 in the labels $V_1(\tau_1, p_1)$, $V_2(\tau_2, p_2)$ below the loop; each vertex carries a schematic fan of two external legs ending in open squares and labeled by the energy sums P_1, P_2. The two massive propagators join the same pair of vertices and are drawn as the two halves of one circular loop — propagator 1 with momentum magnitude x_1 and mass label $\widetilde{\nu}_{B1}$ on the upper half, propagator 2 with x_2 and $\widetilde{\nu}_{B2}$ on the lower half.

From figure 6, the function family of this example has two time integrals, one per vertex, and two propagator kernels, both joining the same pair of vertices; as in the first example each kernel carries one mode-function state at each of its two vertices. Let τ_1, τ_2 be the vertex times, P_1, P_2 the external energy sums, and x_1, x_2 the internal momentum magnitudes. The Hankel orders of the two internal propagators are ν_{B1}, ν_{B2}. This example uses the positive-prefactor states of (2.8),

$$\begin{aligned} h^{(r)}(\nu, 0; z) &= z^{\nu} H_{\nu}^{(r)}(z), \\ h^{(r)}(\nu, 1; z) &= \partial_z h^{(r)}(\nu, 0; z) = z^{\nu} H_{\nu-1}^{(r)}(z), \qquad r = 1, 2. \end{aligned} \tag{4.16}$$

Using these states in the branch products of (4.2) and the SK combinations of (2.3) defines the bubble kernels $\mathcal{G}_{\sigma\sigma',\nu;ab}$. The Top masters are

$$\begin{aligned} \mathrm{I}_{abcd}^{\sigma_1\sigma_2} = \int_{-\infty}^{0} & d\tau_1\, d\tau_2\, (-\tau_1)^{p_1} (-\tau_2)^{p_2} e^{\mathrm{i}(\sigma_1 P_1 \tau_1 + \sigma_2 P_2 \tau_2)} \\ & \times \mathcal{G}_{\sigma_1\sigma_2,\nu_{B1};ab}(x_1; \tau_1, \tau_2) \mathcal{G}_{\sigma_1\sigma_2,\nu_{B2};cd}(x_2; \tau_1, \tau_2). \end{aligned} \tag{4.17}$$

Here $a, b, c, d \in \{0, 1\}$. As in the first example, a, b label the mode-function states of propagator 1 at vertices 1 and 2, and c, d those of propagator 2 in the same vertex order. Here p_1, p_2 are the effective time powers after absorbing the mode prefactors, unlike the bare powers used in (4.6). The contour signs are $\sigma_1, \sigma_2 = \pm 1$. We denote positive and negative signs by p and m, respectively; these labels are unrelated to the time powers p_1, p_2. The four contour assignments are *pp*, *pm*, *mp* and *mm*. An assignment with equal contour signs has 22 masters after the exact mode-function state projection; an assignment with opposite contour signs has only the Top sector, with 16 masters. For each assignment, the

projection onto the Top sector is

$$\begin{aligned}\mathrm{I}^{\sigma_1\sigma_2}_{\mathrm{Top}} = \big(&\mathrm{I}^{\sigma_1\sigma_2}_{0000}, \mathrm{I}^{\sigma_1\sigma_2}_{0001}, \mathrm{I}^{\sigma_1\sigma_2}_{0010}, \mathrm{I}^{\sigma_1\sigma_2}_{0011}, \mathrm{I}^{\sigma_1\sigma_2}_{0100}, \mathrm{I}^{\sigma_1\sigma_2}_{0101}, \mathrm{I}^{\sigma_1\sigma_2}_{0110}, \mathrm{I}^{\sigma_1\sigma_2}_{0111},\\ &\mathrm{I}^{\sigma_1\sigma_2}_{1000}, \mathrm{I}^{\sigma_1\sigma_2}_{1001}, \mathrm{I}^{\sigma_1\sigma_2}_{1010}, \mathrm{I}^{\sigma_1\sigma_2}_{1011}, \mathrm{I}^{\sigma_1\sigma_2}_{1100}, \mathrm{I}^{\sigma_1\sigma_2}_{1101}, \mathrm{I}^{\sigma_1\sigma_2}_{1110}, \mathrm{I}^{\sigma_1\sigma_2}_{1111}\big)^{\mathsf{T}},\end{aligned} \tag{4.18}$$

one for each of the sixteen values of $(a,b,c,d) \in \{0,1\}^4$. The two pinched sectors of an equal-sign assignment arise from contracting one of the two propagators. Both vertices then merge into a single component carrying one time τ, and the surviving propagator joins this component to itself. When its two vertices coincide, the theta pair of the full propagator collapses and its factor becomes the mean of the two chamber products, the spectator factor m_e of (2.70),

$$T_{cd}(x;\tau) = \tfrac{1}{2}\Big[h^{(1)}(\nu,c;-x\tau)\,h^{(2)}(\nu,d;-x\tau) + h^{(2)}(\nu,c;-x\tau)\,h^{(1)}(\nu,d;-x\tau)\Big], \tag{4.19}$$

which is symmetric under $(c,d) \leftrightarrow (d,c)$, so the coincidence quotient identifies $(1,0)$ with $(0,1)$ and keeps the three combinations $(0,0),(0,1),(1,1)$ of the two mode-function states — this is the exact mode-function state projection quoted above. The pinched masters are therefore, with propagator 1 contracted,

$$\mathrm{I}^{\sigma_1\sigma_2,\mathrm{P1}}_{cd} = \int_{-\infty}^{0} \mathrm{d}\tau\,(-\tau)^{p_1+p_2+2\nu_{B1}}\, e^{\mathrm{i}(\sigma_1 P_1+\sigma_2 P_2)\tau}\, T_{cd}(x_2;\tau), \qquad (c,d) \in \{(0,0),(0,1),(1,1)\}, \tag{4.20}$$

where the merged power follows from (2.41) with the raw contact power $\varrho_1 = 2\nu_{B1}-1$ and one vertex merge, and, with propagator 2 contracted,

$$\mathrm{I}^{\sigma_1\sigma_2,\mathrm{P2}}_{ab} = \int_{-\infty}^{0} \mathrm{d}\tau\,(-\tau)^{p_1+p_2+2\nu_{B2}}\, e^{\mathrm{i}(\sigma_1 P_1+\sigma_2 P_2)\tau}\, T_{ab}(x_1;\tau), \qquad (a,b) \in \{(0,0),(0,1),(1,1)\}. \tag{4.21}$$

The equal-sign hierarchy is $\mathrm{Top}(16) \oplus \mathrm{Pinch}_1(3) \oplus \mathrm{Pinch}_2(3)$ with dim = 22 (figure 7).

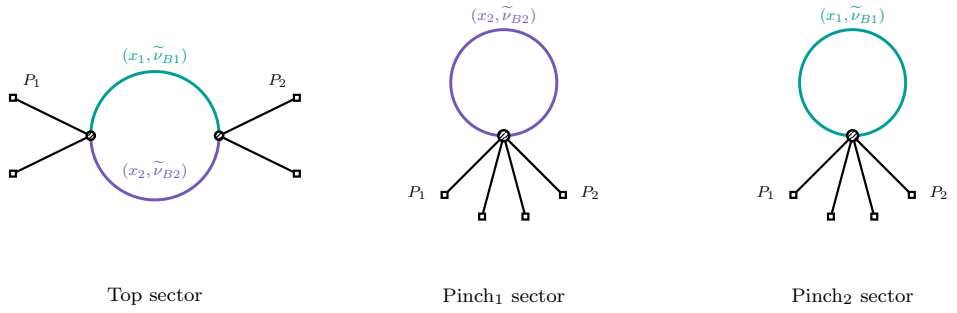


Figure 7. The three sectors of the bubble time kernel of figure 6, drawn for the equal-sign assignments (pp and mm) and with the same colors as in figure 6: the Top sector keeps both propagators of the circular loop; Pinch_1 contracts propagator 1 (upper half, x_1) and leaves propagator 2 as a circular tadpole on the single merged vertex, drawn as a larger hatched circle carrying both external-leg fans; Pinch_2 contracts propagator 2 and leaves propagator 1. The opposite-sign assignments pm and mp have only the Top sector.

The vertex factors $(\mathrm{i}\sigma_1)(\mathrm{i}\sigma_2)$ give

$$\boldsymbol{K}_{\mathrm{SK}} = -\mathrm{I}^{pp}_{\mathrm{Top}} + \mathrm{I}^{pm}_{\mathrm{Top}} + \mathrm{I}^{mp}_{\mathrm{Top}} - \mathrm{I}^{mm}_{\mathrm{Top}}. \tag{4.22}$$

Only the equal-sign assignments reach a pinched sector, and there both propagators are of one branch, so they share the contour sign $\sigma_e = \sigma_1 = \sigma_2$ of (2.16). The hidden constants of the three sectors are

$$\mathcal{N}_{\mathrm{Top}} = 1, \qquad \mathcal{N}_{\mathrm{P1}} = -\sigma_e \frac{4\mathrm{i}}{\pi} x_1^{2\nu_{B1}-1}, \qquad \mathcal{N}_{\mathrm{P2}} = -\sigma_e \frac{4\mathrm{i}}{\pi} x_2^{2\nu_{B2}-1}, \tag{4.23}$$

with $\mathcal{N}_s^{(0)} = 1$; the propagated objects are $J^{\sigma_1\sigma_2}_{abcd} = \mathrm{I}^{\sigma_1\sigma_2}_{abcd}$, $J^{\sigma_1\sigma_2,\mathrm{P1}}_{cd} = \mathcal{N}_{\mathrm{P1}}\mathrm{I}^{\sigma_1\sigma_2,\mathrm{P1}}_{cd}$ and $J^{\sigma_1\sigma_2,\mathrm{P2}}_{ab} = \mathcal{N}_{\mathrm{P2}}\mathrm{I}^{\sigma_1\sigma_2,\mathrm{P2}}_{ab}$. This example runs in the package default convention, so the power is the $2\nu_e - 1$ of (2.16) rather than the $-2\nu_e - 1$ of the tree example. We use $k_s = x_1$ as the momentum unit and vary the dimensionless energy $\rho = P_1/k_s = P_2/k_s$. The parameter values in (4.17) are

$$\begin{aligned} (k_s, x_1, x_2) &= (1, 1, \sqrt{2}), \qquad & P_1 &= P_2 = \rho, \\ (\nu_{B1}, \nu_{B2}) &= (\mathrm{i}/4, \mathrm{i}/4), \qquad & p_1 &= p_2 = 1 + \mathrm{i}/2. \end{aligned} \tag{4.24}$$

The normalized 0000 entry of $\boldsymbol{K}_{\mathrm{SK}}$, written $\widehat{K}_{\mathrm{SK}}(\rho)$, is

$$\widehat{K}_{\mathrm{SK}}(\rho) = \frac{[\boldsymbol{K}_{\mathrm{SK}}(\rho)]_{0000}}{[\boldsymbol{K}_{\mathrm{SK}}(10)]_{0000}}. \tag{4.25}$$

The four zeros select the four undifferentiated mode-function states of the two massive propagators.

The scan contains 201 points over $2.8 \le \rho \le 10$. Calculations use decimal arithmetic with 80 digits. The transport orders are 80 and 96, and the boundary series orders are 17 and 15. The calculation time excluding reference transport at order 96 is 678.20 s. It includes all four contour assignments, DE construction and both boundary calculations. The reference transport takes 5.24 s, and the total recorded time is 683.43 s. The average cost for the scan, including setup, is 3.37 s per point. This average is not the cost of a calculation at a single point.

The maximum relative difference between boundary orders 17 and 15 is 4.740×10^{-24}. The maximum relative difference between transport orders 80 and 96 is 1.025×10^{-50}. These tests assess boundary and transport stability separately; the comparison of transport orders alone is not a bound on the total error.

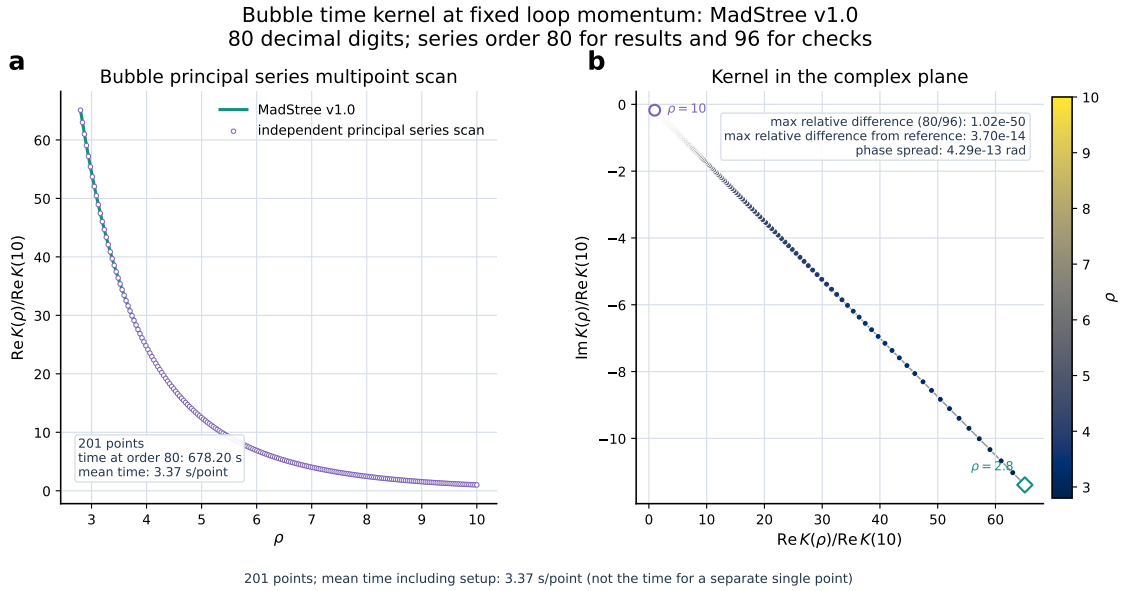


Figure 8. Bubble time kernel evaluated with `MadStree`. **a**, The normalized kernel and the independent reference. **b**, The kernel in the complex plane, with color indicating ρ.

Both panels of figure 8 use the same 201 points. Panel **a** compares the normalized kernel (4.25) with the same quantity computed by a separate program written for this bubble graph alone, which does not use `MadStree`; the maximum relative difference over the scan is 3.70×10^{-14}. The nearly radial distribution in panel **b** indicates an approximately constant phase.

Massless three-vertex tree in the regulator mode The third example returns to the chain topology of the first example with both exchanges massless, and lowers the time power of vertex 1 by two units before shifting all vertex time powers by the common analytic regulator (figure 9): $a_1 = -1+\varepsilon$, $a_2 = a_3 = 1+\varepsilon$. At $\varepsilon = 0$ the time integrals diverge. The regulator mode of Sec. 3.4 evaluates them at the regularized powers; requesting the return level ε^0 delivers both the ε^{-1} pole coefficient and the finite part. The momenta are fixed by `ParameterRules` and the target point is $\{k_1, k_2, k_3\} = \{9\mathrm{i}, 3\mathrm{i}, 5\mathrm{i}\}$.

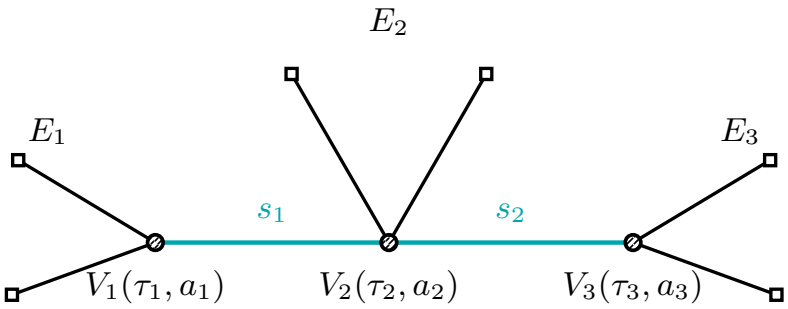


Figure 9. Massless three-vertex chain in the regulator mode, drawn as in figure 2 with the massless counterparts of its parameters: the same external-leg fans with summed energies $E_v = \sum_j k_{v,j}$ and propagator momentum magnitudes s_1, s_2, with no mass labels since the massless propagators carry the fixed order $\nu = 1/2$. The vertex labels $V_v(\tau_v, a_v)$ below the vertices carry the regularized time powers $a_1 = -1 + \varepsilon$ and $a_2 = a_3 = 1 + \varepsilon$, i.e. the time factors $(-\tau_1)^{-1+\varepsilon}$, $(-\tau_2)^{1+\varepsilon}$ and $(-\tau_3)^{1+\varepsilon}$. The fans are schematic — in the input of Example 06 each vertex carries a single external leg, $E_v = k_v$, $s_1 = q_{12}$ and $s_2 = q_{23}$ — and the input fixes all three vertices to the `"+"` branch, so this example evaluates the all-plus assignment $(+\,+\,+)$ only.

The input is

```
treeSpec = <|
  "vertices" -> {
    <|"id" -> 1, "externalLegEnergy" -> k1,
      "timePower" -> a1, "vertexType" -> "+"|>,
    <|"id" -> 2, "externalLegEnergy" -> k2,
      "timePower" -> a2, "vertexType" -> "+"|>,
    <|"id" -> 3, "externalLegEnergy" -> k3,
      "timePower" -> a3, "vertexType" -> "+"|>},
  "lines" -> {
    <|"type" -> "massless", "endpoints" -> {1, 2},
      "momentum" -> q12, "nu" -> 1/2|>,
    <|"type" -> "massless", "endpoints" -> {2, 3},
      "momentum" -> q23, "nu" -> 1/2|>}|>;
context = MSInitTree[treeSpec];
epSeries = MSReconstructEpSeries[context, ep,
  {{k1, k2, k3}, {9 I, 3 I, 5 I}},
  ParameterRules -> {q12 -> 1, q23 -> 2,
    a1 -> -1 + ep, a2 -> 1 + ep, a3 -> 1 + ep},
  MaximumEpPower -> 0, EpGoalDigits -> 12,
  EpSampleAngleRange -> {-Pi/3, Pi/3},
  RankOrder -> {1, 2, 3}]
```

The example additionally sets `BoundaryScale -> 4`, `BoundarySeriesOrder -> 24` and the default pool of `ParallelTaskCount -> 12` workers.

From figure 9, the family has three time integrals and two propagator kernels; because both propagators are massless full propagators of the fixed order $\nu = 1/2$, each contributes one shared mode-function two-state slot (Sec. 2.2) rather than two independent mode-function state slots. We write $E_v = k_v$ for the input external energy and $s_1 = q_{12}$, $s_2 = q_{23}$ for the internal momentum magnitudes. The all-plus input has the vertex phase $e^{-\mathrm{i}E_v\tau_v}$, which damps at the positive-imaginary target energies. At $\nu = 1/2$ the positive-prefactor branches are pure exponentials, so the input powers a_v are the effective integrand powers. We define the integrals in this example directly in the unit-amplitude package basis of (2.28). For a full propagator with contour sign $\sigma = \pm 1$ and shared bit $b \in \{0, 1\}$, its kernel is

$$\mathcal{G}^{\mathrm{m}}_{\sigma,b}(s;\tau_u,\tau_v) = (-1)^b\theta_{uv}e^{-\mathrm{i}\sigma s(\tau_u-\tau_v)} + \theta_{vu}e^{+\mathrm{i}\sigma s(\tau_u-\tau_v)}. \tag{4.26}$$

The two entries are the chamber sum and the second chamber minus the first; the factors $\pi/2$ and $\mathrm{i}\sigma$ relating them to the raw Hankel states are already included. Here $\boldsymbol{\sigma} = (+1, +1, +1)$ and both propagators have $\sigma = +1$. With one shared slot per propagator, the Top sector has the four masters

$$\begin{aligned}\mathrm{I}^{\boldsymbol{\sigma}}_{b_1b_2} = &\int_{-\infty}^{0} \mathrm{d}\tau_1\,\mathrm{d}\tau_2\,\mathrm{d}\tau_3\,(-\tau_1)^{a_1}(-\tau_2)^{a_2}(-\tau_3)^{a_3}e^{-\mathrm{i}(E_1\tau_1+E_2\tau_2+E_3\tau_3)}\\ &\times \mathcal{G}^{\mathrm{m}}_{+1,b_1}(s_1;\tau_1,\tau_2)\,\mathcal{G}^{\mathrm{m}}_{+1,b_2}(s_2;\tau_2,\tau_3), \qquad b_1, b_2 \in \{0, 1\},\end{aligned} \tag{4.27}$$

with the bits taken in propagator order, and the Top vector is $(\mathrm{I}^{\boldsymbol{\sigma}}_{00}, \mathrm{I}^{\boldsymbol{\sigma}}_{01}, \mathrm{I}^{\boldsymbol{\sigma}}_{10}, \mathrm{I}^{\boldsymbol{\sigma}}_{11})^{\mathsf{T}}$. A contracted massless propagator has the raw contact power $\varrho = 0$ of (2.41), and its pure exponentials multiply into the merged exponential. The four sectors, shown in figure 3, are

$$\begin{aligned}\mathrm{I}^{\boldsymbol{\sigma},\mathrm{L}}_{b} = &\int_{-\infty}^{0} \mathrm{d}\tau_{12}\int_{-\infty}^{0} \mathrm{d}\tau_3\,(-\tau_{12})^{a_1+a_2+1}(-\tau_3)^{a_3}e^{-\mathrm{i}(E_1+E_2)\tau_{12}-\mathrm{i}E_3\tau_3}\\ &\times \mathcal{G}^{\mathrm{m}}_{+1,b}(s_2;\tau_{12},\tau_3), \qquad b \in \{0, 1\},\end{aligned} \tag{4.28}$$

where propagator 1 is contracted and τ_{12} is the time of the merged component,

$$\begin{aligned}\mathrm{I}^{\boldsymbol{\sigma},\mathrm{R}}_{b} = &\int_{-\infty}^{0} \mathrm{d}\tau_1\int_{-\infty}^{0} \mathrm{d}\tau_{23}\,(-\tau_1)^{a_1}(-\tau_{23})^{a_2+a_3+1}e^{-\mathrm{i}E_1\tau_1-\mathrm{i}(E_2+E_3)\tau_{23}}\\ &\times \mathcal{G}^{\mathrm{m}}_{+1,b}(s_1;\tau_1,\tau_{23}), \qquad b \in \{0, 1\},\end{aligned} \tag{4.29}$$

with propagator 2 contracted, and the single one-time master of the DoublePinch sector,

$$\mathrm{I}^{\boldsymbol{\sigma},\mathrm{D}} = \int_{-\infty}^{0} \mathrm{d}\tau\,(-\tau)^{a_1+a_2+a_3+2}e^{-\mathrm{i}(E_1+E_2+E_3)\tau}. \tag{4.30}$$

The hierarchy is therefore Top(4) $\oplus$ LeftPinch(2) $\oplus$ RightPinch(2) $\oplus$ DoublePinch(1) with dim = 9, the nine-dimensional DE recorded in Sec. 4.1; the massive chain of (4.10) carries four mode-function state slots and 25 masters. No contracted propagator contributes a pinch factor in (2.16), so every sector normalization is 1. Since the integrals in this example are already defined in the package basis, $J_s = \mathrm{I}_s$ here, and the propagated vector is I itself,

without vertex contour factors.

Before the regulator enters, the initialized context is already a closed reduction system, and `MSReduce` accepts shifted combinations of the integrals just defined. Taking the two integrals whose time powers lie one unit below the master powers, at vertex 1 and at vertex 2, and reducing their sum in one call,

```
combination = MSIntegral["11", {-1, 0, 0}, {0, 0}] +
              MSIntegral["11", {0, -1, 0}, {1, 0}];
reduction = MSReduce[combination, context];
```

returns `"status" -> "reduced"` with a zero `"nonMasterResidual"` and an empty `"remainingShiftedIntegrals"`. Writing $\mathrm{I}^{\boldsymbol{\sigma}}_{b_1 b_2}\big|_{a_v \to a_v - 1}$ for the integral with the time power a_v of (4.27) lowered by one unit,

$$\begin{aligned}\mathrm{I}^{\boldsymbol{\sigma}}_{00}\big|_{a_1\to a_1-1} + \mathrm{I}^{\boldsymbol{\sigma}}_{10}\big|_{a_2\to a_2-1} = &-\mathrm{i}\,\frac{a_2E_1 + a_1 s_1}{a_1 a_2}\,\mathrm{I}^{\boldsymbol{\sigma}}_{00} + \mathrm{i}\left(\frac{s_1}{a_1} - \frac{E_2}{a_2}\right)\mathrm{I}^{\boldsymbol{\sigma}}_{10} + \mathrm{i}\,\frac{s_2}{a_2}\,\mathrm{I}^{\boldsymbol{\sigma}}_{11} \\ &- \frac{2\mathrm{i}\,(E_1+E_2)}{a_2(1+a_1+a_2)}\,\mathrm{I}^{\boldsymbol{\sigma},\mathrm{L}}_{0} + \frac{2\mathrm{i}\,s_2}{a_2(1+a_1+a_2)}\,\mathrm{I}^{\boldsymbol{\sigma},\mathrm{L}}_{1}.\end{aligned} \tag{4.31}$$

The lowering directions invert M_1 eigenvalues, the contact of the first lowering injects into the LeftPinch sector, whose merged component is lowered once more, and the returned `"singularLayers"` names exactly the inverted denominators $\{a_1\}$, $\{a_2\}$ and $\{1+a_1+a_2\}$. The reduction uses the time powers fixed at initialization, so substituted values are entered by initializing with them: with the regularized powers $a_1 = -1+\varepsilon$, $a_2 = a_3 = 1+\varepsilon$ in the `timePower` fields of the same topology, the same call returns the LeftPinch coefficient $-2\mathrm{i}(E_1+E_2)/[(1+\varepsilon)(1+2\varepsilon)]$ and the singular surfaces $\{-1+\varepsilon\}$, $\{1+\varepsilon\}$, $\{1+2\varepsilon\}$, exposing the ε-pole that the Laurent reconstruction below resolves.

The certificate of Sec. 3.4 runs before any numerical solve and returns $n_{\min} = -1$: the d log-form DE is regular at $\varepsilon = 0$, but the vertex-1 unit $u_1 = a_1 + 1 = \varepsilon$ is isolated as a standalone factor in part of the factorized boundary weights, and every such branch carries a single $\Gamma(\varepsilon)$ simple pole, so the combined physical boundary starts at ε^{-1} with component valuations -1 or 0 only. Three Frobenius recurrence operators do drop rank at $\varepsilon = 0$, with rank changes $9 \to 5$, $9 \to 8$ and $9 \to 8$ at the exponents $6(1+\varepsilon)$ (twice) and $8 + 7\varepsilon$; per Sec. 3.4 these are retained as resonance diagnostics and produce no additional pole, since the combined physical boundary decides the support. With $m = 0$, $n_{\min} = -1$ and $g = 12$, (3.3) and (3.4) give $P = 1$, $\Delta = 1$, $N_{\mathrm{fit}} = 4$, $\alpha_\varepsilon = 25/4$, $\varepsilon_0 = 10^{-25/4}$, $p_0 = 32$, a working precision of 200 digits and a primary transport order of 128 with 50 reference orders. The four production points are exact Gaussian rationals of moduli $\varepsilon_0(1 + j/100)$, $j = 1, \ldots, 4$, cycling through the rays $-\pi/6$, 0 and $+\pi/6$ of the open window $(-\pi/3, \pi/3)$; the two validation points sit at half the first two moduli, on the rays $-\pi/6$ and 0, and never enter the fit. All six solves are dispatched to one bounded pool (12 requested, 6 effective); the whole reconstruction, including the certificate, both validation solves and the fit, takes 170 s of wall time.

The returned Laurent data are the ε^{-1} pole coefficient and the ε^0 coefficient, both 9-entry master vectors. Entries below 10^{-20} in absolute value are numerical noise at the

achieved accuracy and are reported as zero. For scale, the relative validation residual is of order 10^{-25}, while the physical entries are of order 10^{-3} or larger. With this convention three entries of the pole coefficient vanish, among them the DoublePinch entry, because the fully merged boundary branch carries no isolated $\Gamma(\varepsilon)$ factor. Two representative entries are

$$\begin{aligned}[c_{-1}]_{\mathrm{Top},00} &= \left(9.49200055589 - 25.86650516658\,\mathrm{i}\right) \times 10^{-4}, \\ [c_0]_{\mathrm{D}} &= \frac{\Gamma(4)}{17^4} = 7.183822032782174 \times 10^{-5},\end{aligned} \tag{4.32}$$

The first is the Top-sector entry with both shared slots in their symmetric state, $b_1 = b_2 = 0$, and the second is the DoublePinch master, a single one-time integral. At the two independent validation points the fitted series is summed over all four internal powers $\varepsilon^{-1}, \ldots, \varepsilon^2$, which include the two buffer powers above the requested maximum $m = 0$. Compared with the directly transported values it reaches a maximum relative residual of 7.43×10^{-25}, thirteen orders below the requested tolerance 10^{-12}. That residual certifies the fit against this tolerance, not a number of correct digits: the coefficients are computed at a working precision of 200 digits and returned with about 199 significant digits.

The tree example illustrates two uses of DE transport: efficient multipoint evaluation and continuation beyond the convergence domain of a direct series. The bubble example applies the same framework for time integrals to a graph containing a loop at fixed internal momenta. The regulator example certifies the Laurent support symbolically on a divergent massless chain and delivers the $\varepsilon \to 0$ pole coefficient and finite part from six numerical solves.

5 Verification against independent references

The package was checked against independent references in the following verification tasks.

- **Single-vertex families and the massive two-vertex correlator.** The exact differential equations, the boundary coefficients and the numerically transported results are compared with the known results of [43], which are computed by an independent multivariate hypergeometric construction.

- **The massive three-vertex tree graph.** The package results are compared with the published eight-branch result of [3] and an independent three-vertex implementation generated by an agent from the hand-written two-vertex code. This reference implementation was first checked against the published result, then used to validate `MadStree`.

- **IBP relations and differential equations.** These are compared with the output of the `dSIBP` package [63], which is released in the same repository as this one, https://github.com/J7C/dSIBP-Packages. `dSIBP` samples naive IBP seeds and solves the resulting linear reduction system, whereas the present package applies the general formulas derived in Sec. 2 directly and never re-solves an IBP system; the agreement therefore tests the closed formulas themselves.

6 Summary and outlook

`MadStree` provides closed IBP reduction and automated numerical evaluation for arbitrary dS tree families: every convergent integral of a specified family reduces to its master basis through explicit recurrence formulas, and the same reduction covers the time integrals of loop graphs at fixed internal momenta. The master representation is kept small by the extra relations of a massless full propagator, which replace four combinations of its two mode-function states by one shared two-state factor. Applying these relations to the general construction of [42] gives closed IBP recurrences and a d log-form DE on the smaller space. The generic Hankel reduction and DE formulas remain applicable without this compression, but retain redundant states. For the equal-sign three-vertex chain, the compressed massless hierarchy has nine masters, compared with the 25 states of the generic hierarchy.

For a specified tree and contour assignment, the package constructs the sectors, recurrence matrices, connection and boundary data, and reduces any integral of the family on request, without requiring intermediate formulas from the user. The boundary construction extends ordered large-energy limits to general tree topology, including vertices without external legs through auxiliary energies. Numerical transport with `FlintNDE` then supplies values along a kinematic path. The massive-tree benchmark illustrates how sharing boundary data and local series lowers the cost per output point in a scan, and how transport continues the selected branch beyond the convergence domain of a direct hypergeometric series. Laurent reconstruction additionally returns pole coefficients and finite parts for regularized time integrals.

These calculations apply to the scalar integral families defined in Sec. 2.1. Physical correlators require the appropriate contour sum and normalization factors, as illustrated in Sec. 4.2. Symbolic checks constrain the connection, boundary charts and Laurent support; comparisons of truncation orders and independent values assess numerical stability. Their residuals are error estimates rather than general bounds on the final error, and the reported status and precision fields must be inspected.

The common-theta construction also permits time integrations for graphs with cycles at fixed internal momenta. Spatial loop-momentum integration remains outside the package. The loop-level IBP and d log-form construction of [62] provides an algebraic basis for extending this workflow; incorporating the momentum integrations and their additional sectors into an automated numerical calculation is a further step.

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Statement on AI Use. The authors designed the package notation, user-facing functions and input formats, and set the overall direction and strategy of the derivation in Sec. 2.

Several agent programs, each driven by a different large language model, worked out the derivation and implemented the package based on its results and the authors' guidance.

A Common-theta bundles and the odd-subset expansion

Bundles and their theta argument. Fix a sector s and write $\mathrm{comp}_s(v)$ for the current component that contains the vertex v; a component is a maximal set of vertices already merged by contacts, and a lone vertex is one component. An uncontracted full propagator $e = (u_e, v_e)$ carries the theta $\theta(\tau_{u_e} - \tau_{v_e})$ of its two component times, so two propagators e, e' carry the *same* theta if and only if their component pairs coincide as ordered pairs,

$$\begin{gathered}\big(\mathrm{comp}_s(u_e), \mathrm{comp}_s(v_e)\big) = \big(\mathrm{comp}_s(u_{e'}), \mathrm{comp}_s(v_{e'})\big) = (U, V), \\ \text{so} \quad \Delta_e = \Delta_{e'} \equiv \Delta = \tau_U - \tau_V .\end{gathered} \tag{A.1}$$

A maximal set of propagators with this property is the common-theta bundle $\mathcal{B}$ of the pair (U, V), with $|\mathcal{B}| \geq 2$. The steps below also cover $|\mathcal{B}| = 1$, which is the ordinary single-propagator contact treated in Sec. 2.4. Different component pairs define different bundles. A time derivative acts on their theta functions by the product rule, so each contact term is assigned to the bundle whose theta was differentiated. The vertex exponentials of Block (i) are theta-free and take no part in this bookkeeping, so only the mode-function two-state factors of (2.3) enter the steps below.

Step 1: multiplying the thetas out. For each propagator of one bundle, write the SK pair of products as in (2.3), α_e for the coefficient of $\theta(\Delta)$ and β_e for that of $\theta(-\Delta)$. At the level of the h blocks the two are the two orderings of the Hankel branches,

$$\alpha_e = h^{(1)}_{u_e}\, h^{(2)}_{v_e}, \qquad \beta_e = h^{(2)}_{u_e}\, h^{(1)}_{v_e}, \tag{A.2}$$

where the orders, mode-function states and arguments are those of (2.8); G_{--} interchanges the two names, and this is the only place the sign σ_e of the bundle enters. The product of the bundle is a product of $|\mathcal{B}|$ binomials in the two functions $\theta(\Delta)$ and $\theta(-\Delta)$. Expanding naively gives $2^{|\mathcal{B}|}$ terms, but inside one bundle $\theta(\Delta)$ is the *same* function for every factor, so idempotency $\theta(\Delta)^2 = \theta(\Delta)$ makes all powers of one argument collapse to that argument, and orthogonality $\theta(\Delta)\theta(-\Delta) = 0$ annihilates every term containing both arguments. Exactly the two terms of (2.69) survive. The factor $2^{|\mathcal{B}|}$ thus never appears within a bundle: it would count the events of $|\mathcal{B}|$ propagators with pairwise different theta arguments, that is $|\mathcal{B}|$ bundles of one propagator each, and those are enumerated bundle by bundle.

Step 2: one theta gives one delta. The contact part of the Δ derivative of the two-term form comes from differentiating only the step functions, leaving the coefficients of (A.2) unchanged:

$$\big(\partial_\Delta \theta(\Delta)\big) \prod_{e\in\mathcal{B}} \alpha_e + \big(\partial_\Delta \theta(-\Delta)\big) \prod_{e\in\mathcal{B}} \beta_e = \delta(\Delta) \left(\prod_{e\in\mathcal{B}} \alpha_e - \prod_{e\in\mathcal{B}} \beta_e \right), \tag{A.3}$$

because $\partial_\Delta\theta(\Delta)=\delta(\Delta)$ and $\partial_\Delta\theta(-\Delta)=-\delta(\Delta)$. The coefficients α_e and β_e are products of mode functions of the two component times and so depend on Δ as well; the complete derivative therefore adds the two terms $\theta(\Delta)\,\partial_\Delta\prod_e\alpha_e$ and $\theta(-\Delta)\,\partial_\Delta\prod_e\beta_e$, and those are the regular derivative terms carried by the M_0 and M_1 generators of Sec. 2.2; they force no two component times to coincide. Only the contact part (A.3) enters the bookkeeping of this appendix. Three consequences are worth stating explicitly. First, the bundle produces *one* distribution, not $|\mathcal{B}|$ of them: the whole contact information of the bundle is concentrated in the single coefficient $\prod_e\alpha_e-\prod_e\beta_e$, which is antisymmetric under $\alpha\leftrightarrow\beta$. Second, that one $\delta(\Delta)$ forces the two component times to coincide *once*, so the components U and V merge once, independently of how many factors d_e a term of the coefficient contains; this is what fixes the merge count in (2.41). Third, for G_{--} the interchange of α and β turns (A.3) into its negative, which reproduces the single sign σ_e of (2.27) for a whole bundle.

Step 3: telescoping the antisymmetric difference. Order the propagators of the bundle as $1,\dots,|\mathcal{B}|$ and abbreviate $\alpha_e=A_e$, $\beta_e=B_e$. Inserting and subtracting successive partial products gives the telescoping identity

$$\begin{aligned}\prod_{e=1}^{|\mathcal{B}|}A_e-\prod_{e=1}^{|\mathcal{B}|}B_e&=\sum_{k=1}^{|\mathcal{B}|}\left(\prod_{e\le k}A_e\prod_{e>k}B_e-\prod_{e<k}A_e\prod_{e\ge k}B_e\right)\\&=\sum_{k=1}^{|\mathcal{B}|}D_k\prod_{e<k}A_e\prod_{e>k}B_e,\end{aligned}\tag{A.4}$$

whose intermediate sum collapses because the second summand of the k-th term equals the first summand of the $(k{+}1)$-th term, while the two ends, $k=1$ and $k=|\mathcal{B}|$, supply the two products of (A.3). Each term of (A.4) contains exactly one difference factor $D_e=A_e-B_e$.

Step 4: collecting by subsets and the parity. Introduce, as in (2.70), the difference and the mean of the two products, $\mathsf{d}_e=D_e=A_e-B_e$ and $\mathsf{m}_e=(A_e+B_e)/2$, equivalently $A_e=\mathsf{m}_e+\mathsf{d}_e/2$ and $B_e=\mathsf{m}_e-\mathsf{d}_e/2$, and expand the spectators of (A.4) in this basis. In the k-th term the propagator k supplies the explicit d_k; a propagator $e\in S$ with $e<k$ contributes $+\mathsf{d}_e/2$, a propagator $e\in S$ with $e>k$ contributes $-\mathsf{d}_e/2$, and every propagator outside S contributes m_e. So the coefficient of the monomial $\prod_{e\in S}\mathsf{d}_e\prod_{e\notin S}\mathsf{m}_e$ is

$$c_S=2^{1-|S|}\sum_{k\in S}(-1)^{\#\{e\in S:\,e>k\}}.\tag{A.5}$$

Writing $S=\{i_1<\cdots<i_m\}$ with $m=|S|$, the exponent for $k=i_j$ is $m-j$, hence

$$\sum_{j=1}^{m}(-1)^{m-j}=\sum_{r=0}^{m-1}(-1)^r=\begin{cases}1, & m\text{ odd},\\ 0, & m\text{ even},\end{cases}\qquad c_S=\big(1-(-1)^{|S|}\big)2^{-|S|},\tag{A.6}$$

which is (2.70): only odd subsets survive, each with the pure numerical weight $2^{1-|S|}$. The empty set cannot occur, since every telescoping term carries at least the explicit d_k, in

agreement with $c_\varnothing = 0$. The same parity follows from symmetry without any enumeration: the left-hand side of (A.4) changes sign under $\alpha \leftrightarrow \beta$, in which d_e is odd and m_e even, so a monomial with an even number of d factors must cancel; the explicit computation (A.5) only fixes the remaining coefficients.

Events, weights and the bookkeeping. Each non-empty odd subset S of one bundle is one simultaneous contact event. Its injection matrix $C_{st}^{I,(v,1)}$ of (2.50) is the tensor product of the local contact selectors of the propagators in S — $|1\rangle_e$ on a massless shared slot, the complementary-bit selector of (2.49) on a pair of massive-propagator mode-function states — including their raw Wronskian weights, multiplied by $2^{1-|S|}$ and by the spectator factors m_e of the unselected bundle propagators, then embedded in the global bit order with all spectator bits of other propagators preserved. The collected form (2.70), not the term-by-term form (A.4), is what the package enumerates: the k-th telescoping term stands for "contract only the k-th propagator," but after the spectators are expanded the same subset S is reached from several k, so an enumeration over k would generate the same sector transition more than once. After collecting, each event appears exactly once with a numerical weight, and the deduplication key of the reduction is the triple (parent sector, child sector, set of selected propagator identifiers). There are only finitely many such events, because every event strictly reduces the number of current components.

Child base power and coincident unselected propagators. For a component C produced by such events the base power is (2.41), $A_{C,s} = \sum_{v\in C} \nu_{0,v} + \sum_{e\in E_s(C)} \varrho_e + |C| - 1$. The vertex powers add because the merged integrand is a product. Each contracted propagator contributes its raw contact power ϱ_e through the factor $(-\tau)^{\varrho_e}$ accompanying the corresponding δ. The last term counts component merges: $|C| - 1$ for $|C|$ vertices, regardless of how many bundle propagators are contracted at once. Since (2.41) depends only on the final component and on the contracted set, and not on the order or the clustering of the events, different legal event sequences that reach the same sector give the same child base power; the reduction is confluent at the level of the powers.

An unselected propagator whose two vertices fall into one component after a bundle event must also be reduced with the equal-time identifications. For a massless full propagator the shared slot then contributes only its symmetric mode-function state, because the two Wightman combinations are equal at equal times, so the antisymmetric shared mode-function state of the propagator gives zero; for a massive full propagator the two mode-function states carrying a single derivative become identified, $10 \sim 01$, which is the equal-time canonical quotient. Both cases are implemented the same way: the matrices are first built in the raw binary space of all massive and shared slots and then retracted to the true sector masters by the constant maps of (2.24), $P_s\, M^{\mathrm{raw}}\, S_s$, so the equal-time quotients require no further computation and no dense inversion. On the boundary side the same equal-time configuration makes the two branch arrangements of a coincident massive full propagator enter with the same power, and the product of its two mode-function asymptotic coefficients is averaged over these two Hankel-branch arrangements, one mean per coincident propagator, as stated in Sec. 2.6.

Relation to the single-contact case. For $|\mathcal{B}| = 1$ the two-term form (2.69) reduces to the ordinary SK pair of one propagator, (A.4) to the single difference d_e, and $c_{\{e\}} = 1$, so the bundle machinery reproduces the single-propagator contacts of Sec. 2.4 and of [42]. The new ingredients are the weight $2^{1-|S|}$, the parity selection, and the identification of the merge count with $|C| - 1$ rather than with the number of contracted propagators.

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